

ĐẠI HỌC QUỐC GIA HÀ NỘI
TRƯỜNG ĐẠI HỌC CÔNG NGHỆ



Le Xuan An

**Gradient-Routed Alignment with Capacity-aware Experts
for Multimodal Understanding**

KHOÁ LUẬN TỐT NGHIỆP

Ngành: Khoa học máy tính

Giảng viên hướng dẫn: Tiến sĩ Nguyễn Văn Sơn

Giảng viên đồng hướng dẫn: Phó Giáo sư Võ Đình Hiếu

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VIETNAM NATIONAL UNIVERSITY, HANOI
UNIVERSITY OF ENGINEERING AND TECHNOLOGY



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for Multimodal Understanding**

BACHELOR'S THESIS

Major: Computer Science

Advisor: Dr. Nguyen Van Son

Co-Supervisor: Associate Professor Vo Dinh Hieu

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Lời cảm ơn

Lời đầu tiên, tôi xin gửi lời cảm ơn chân thành và lòng biết ơn sâu sắc tới thầy TS. Nguyễn Văn Sơn và thầy PGS.TS. Võ Đình Hiếu vì đã hướng dẫn và hỗ trợ tôi về kiến thức chuyên môn vô cùng tận tình trong quá trình nghiên cứu và thực hiện khóa luận này.

Tôi cũng xin được gửi lời cảm ơn đến thầy cô, các anh chị và các bạn trong phòng thí nghiệm *iSE - Intelligent Software Engineering* đã luôn hỗ trợ tôi về kiến thức và chuyên môn trong quá trình làm khoá luận.

Cuối cùng, tôi xin gửi lời cảm ơn chân thành tới các thầy, cô trong trường Đại học Công Nghệ - Đại học Quốc gia Hà Nội đã tạo điều kiện cho tôi được học tập và nghiên cứu để có thể hoàn thành khoá luận với kết quả tốt nhất.

Assurance

I hereby affirm that this thesis is my own original work. I have not submitted it previously for any academic or professional award.

I completed this thesis through my own efforts, with guidance and support from Dr. Nguyen Van Son and Associate Professor Vo Dinh Hieu, in partial fulfillment of the graduation requirements. I confirm that all information, data and findings presented here are original and free from plagiarism.

I have acknowledged every source that I consulted and included each one in the references. I accept full responsibility for the accuracy and integrity of this thesis under the regulations of VNU University of Engineering and Technology.

Hanoi, Ngày 17 tháng 5 năm 2026

Student

Le Xuan An

Tóm tắt

Khóa luận này nghiên cứu bài toán chưng cất tri thức đa giáo viên không đồng nhất cho mô hình thị giác-ngôn ngữ (VLM). Mục tiêu là truyền tri thức từ một tập mô hình giáo viên đóng băng sang một mô hình học sinh nhỏ hơn khi các giáo viên thuộc những họ mô hình khác nhau, sử dụng bộ tách từ khác nhau và có mức độ hữu ích thay đổi theo từng mẫu. Trong bối cảnh này, chưng cất trực tiếp theo chỉ số token không còn hợp lệ và phép lấy trung bình cố định giữa giáo viên có thể trộn tín hiệu hữu ích với tín hiệu không liên quan hoặc xung đột.

Để giải quyết bài toán trên, khóa luận đề xuất Gradient-Routed Alignment with Capacity-aware Experts (GRACE), một khung chưng cất cho truyền tri thức VLM đa giáo viên không đồng nhất. GRACE kết hợp ba cơ chế: chưng cất Byte-Trie Wasserstein trên cây byte UTF-8 dùng chung, định tuyến giáo viên thừa top- k từ trạng thái ẩn của mô hình học sinh và tinh chỉnh trọng số bằng độ đồng thuận gradient. Hàm mất mát Byte-Trie Wasserstein giúp so sánh phân phối đầu ra của giáo viên và học sinh giữa các bộ tách từ khác nhau. Định tuyến thừa chọn các giáo viên phù hợp với từng mẫu. Tinh chỉnh bằng độ đồng thuận gradient sau đó điều chỉnh trọng số đã định tuyến theo mức độ căn chỉnh giữa gradient chưng cất của từng giáo viên và gradient cross-entropy có giám sát.

Thực nghiệm trên TextVQA, DocVQA và ChartQA đánh giá GRACE với hai mô hình học sinh SmolVLM-500M và SmolVLM-256M trong thiết lập bốn giáo viên không đồng nhất. Với SmolVLM-500M, GRACE đạt 69.38, 73.11 và 80.20 trên TextVQA, DocVQA và ChartQA, so với điểm gốc 56.75, 59.52 và 62.80 của mô hình học sinh. Các kết quả này tương ứng với mức cải thiện tương đối 22.26%, 22.83% và 27.71% so với mô hình gốc, đồng thời thu hẹp 55.03%, 39.53% và 55.34% khoảng cách ban đầu tới giáo viên có điểm cao nhất. Với SmolVLM-256M, GRACE đạt 60.73, 60.48 và 72.32 trên cùng ba bộ đánh giá. Trên các thực nghiệm được báo cáo, GRACE vượt qua tinh chỉnh trực tiếp, chưng cất logits cùng họ mô hình, ULD khác họ mô hình và các phương pháp đa giáo viên không đồng nhất được so sánh.

Từ khóa: mô hình thị giác-ngôn ngữ, chưng cất tri thức, chưng cất đa giáo viên, định tuyến giáo viên thừa, đồng thuận gradient, chưng cất Wasserstein khác bộ tách từ

Abstract

This thesis studies heterogeneous multi-teacher knowledge distillation for vision-language models (VLMs). The problem is to transfer supervision from a pool of frozen teacher models to a smaller student when the teachers come from different model families, use different tokenizers and vary in usefulness across samples. In this setting, direct token-index distillation is invalid and fixed teacher averaging can mix helpful teacher signals with irrelevant or conflicting ones.

To address this problem, the thesis proposes Gradient-Routed Alignment with Capacity-aware Experts (GRACE), a distillation framework for heterogeneous teacher-to-student VLM transfer. GRACE combines three mechanisms: Byte-Trie Wasserstein knowledge distillation over a shared UTF-8 byte trie, sparse top- k teacher routing from student hidden states and gradient-agreement refinement of routed teacher weights. The Byte-Trie Wasserstein loss makes teacher and student output distributions comparable across tokenizers, while sparse routing and gradient-agreement refinement select teachers relevant to each sample and weight them by CE-gradient alignment.

Experiments on TextVQA, DocVQA and ChartQA evaluate GRACE with SmolVLM-500M and SmolVLM-256M students under a four-teacher heterogeneous pool. For SmolVLM-500M, GRACE reaches 69.38, 73.11 and 80.20 on TextVQA, DocVQA and ChartQA, compared with original student scores of 56.75, 59.52 and 62.80. These results correspond to relative gains of 22.26%, 22.83% and 27.71% over the original student and close 55.03%, 39.53% and 55.34% of the original gap to the highest-scoring teacher. For SmolVLM-256M, GRACE reaches 60.73, 60.48 and 72.32 on the same benchmarks. Across the reported experiments, GRACE improves over direct fine-tuning, same-family logit-based KD, cross-family ULD and the compared heterogeneous multi-teacher baselines.

Keywords: vision-language model, knowledge distillation, multi-teacher distillation, sparse teacher routing, gradient agreement, cross-tokenizer Wasserstein distillation

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List of Abbreviations

Abbr.	Full Term	Description
ANLS	Average Normalized Levenshtein Similarity	Evaluation metric used for DocVQA.
CE	Cross-Entropy	Supervised answer-token loss used during training.
CKA	Centered Kernel Alignment	Representation similarity measure used to compare hidden layers.
ChartQA	Chart Question Answering	Benchmark for question answering over charts.
DocVQA	Document Visual Question Answering	Benchmark for question answering on document images.
EOS	End of Sequence	Special token that marks the end of a generated answer span.
GRACE	Gradient-Routed Alignment with Capacity-aware Experts	Proposed multi-teacher distillation method in this thesis.
SHM-CKA	Hidden-State Matching with CKA	Baseline that combines logits distillation with hidden-state CKA matching.
JS	Jensen-Shannon	Symmetric divergence measure between probability distributions.
KD	Knowledge Distillation	Transfer of supervision from teacher models to a student model.
KL	Kullback-Leibler	Divergence used to compare probability distributions.
LLM	Large Language Model	Language model with large parameter scale and generation capability.
MLP	Multi-Layer Perceptron	Feedforward projection block used in the teacher routing network.
MoE	Mixture of Experts	Sparse routing framework that motivates teacher selection.
MTKD	Multi-Teacher Knowledge Distillation	Distillation setting that uses more than one teacher model.

Abbr.	Full Term	Description
OCR	Optical Character Recognition	Extraction of text from images or scanned documents.
OT	Optimal Transport	Distribution-matching view used to motivate trie-Wasserstein KD.
TWD	Tree-Wasserstein Distance	Tree-structured optimal-transport distance used as background for the proposed loss.
TextVQA	Text Visual Question Answering	Benchmark for question answering on natural images containing text.
ULD	Universal Logit Distillation	Tokenizer-agnostic distillation baseline based on ranked probability distributions.
VLM	Vision-Language Model	Model that processes visual and textual inputs together.
VLM-Eval-Kit	Vision-Language Model Evaluation Kit	Toolkit used to run standardized multimodal evaluation.
VQA	Visual Question Answering	Task of answering questions grounded in visual inputs.

Chapter 1

Introduction

Motivation

Vision-language models (VLMs) link a vision encoder, a multimodal connector and an autoregressive language decoder to read text embedded in images, interpret document layouts, parse charts and answer questions about visual scenes [1–6].

Neural scaling laws push the most capable VLMs toward larger language backbones and more extensive multimodal training, consistently improving performance at the cost of compute and memory [7, 8]. Current frontier models make this tension concrete: Qwen2.5-VL-72B, NVLM-D 1.0 72B, Meta’s Llama 3.2 Vision 90B and Llama 3-V 405B represent the upper range of open-weight and open-access VLMs [9–12]. In bfloat16 inference, storing weights alone requires roughly $2P$ bytes for P parameters, placing these models at approximately 144 GB, 180 GB and 810 GB before key-value cache, activations and runtime overhead, well above the 80 GB of a single H100 [13]. Serving them therefore requires multi-GPU configurations, raising latency, energy cost and infrastructure complexity. These deployment barriers extend beyond research: robotics, embodied AI, mobile applications, document-processing pipelines and enterprise systems all impose real-time, cost and security constraints that the largest models cannot meet [14, 15].

Knowledge distillation addresses this tension by transferring knowledge from a large teacher model to a compact student. Rather than learning from raw annotations alone, the student is trained against the teacher’s output distributions and intermediate representations, which carry soft probability signals over alternative answers and provide richer supervision than one-hot labels [16, 17]. Distillation has produced compact models that retain most of their teachers’ accuracy, with recent lightweight VLMs reaching competitive benchmark scores under resource constraints [18–20].

Research Gaps and Proposed Approach

No single approach resolves all of these constraints simultaneously. Response-based distillation aligns teacher and student output distributions effectively but requires a shared tokenizer [16, 17]. Transport-based methods remove this constraint by comparing prob-

ability mass over decoded strings, yet they are typically formulated for single-teacher transfer, leaving sample-level teacher selection outside their scope [21–23]. Representation matching through hidden states or CKA provides a tokenizer-independent signal but does not calibrate answer probabilities or resolve conflicts between teachers [24]. Multi-teacher distillation improves by pooling supervision from several teachers; however, fixed or globally learned weights can mix helpful teacher outputs with weak or conflicting ones at the sample level [25–27].

These gaps are most acute for text-heavy visual tasks, including document QA, chart interpretation and scene-text understanding, where teachers from different families tokenize the same answer strings differently and complement each other across task types. Although recent open-source VLMs have narrowed the gap to proprietary systems on standard benchmarks [28–30], compact models still struggle when fine visual details, document structure and numerical relations must all be captured. To address these constraints, this thesis proposes and evaluates **Gradient-Routed Alignment with Capacity-aware Experts (GRACE)**, a heterogeneous multi-teacher VLM distillation framework that combines **tokenizer-agnostic** distribution matching, **sample-dependent** teacher selection and **gradient-aware conflict resolution**. Three research questions guide the evaluation:

- **RQ1: Performance.** How does GRACE compare with direct fine-tuning, same-family logit-based KD, ULD and heterogeneous multi-teacher baselines on the target VQA benchmarks?
- **RQ2: Component Analysis.** How much do trie-Wasserstein KD, sparse top- k routing and gradient-agreement refinement contribute to the final performance?
- **RQ3: Efficiency.** What runtime overhead does GRACE introduce during training?

Contributions

This thesis makes three main contributions:

- **Byte-Trie Wasserstein distillation.** We propose a Byte-Trie Wasserstein KD loss that maps teacher and student tokens onto a shared UTF-8 byte trie, comparing byte paths rather than vocabulary indices to resolve tokenizer mismatch.
- **Objective-aware multi-teacher weighting.** We introduce a mechanism for weighting routed teachers where top- k routing selects candidate teachers and gradient agreement filters their KD signals against the supervised CE objective.
- **Empirical validation on VQA benchmarks.** We evaluate the full framework on TextVQA, DocVQA and ChartQA with two SmolVLM students and a heterogeneous four-teacher pool. The results show that GRACE outperforms fine-tuning, same-family KD, cross-family ULD and multi-teacher baselines, while ablations show that routing is unreliable until gradient-agreement refinement is added.

Evaluated with VLM-Eval-Kit on TextVQA, DocVQA and ChartQA, GRACE with SmolVLM-500M as student closes 55%, 40% and 55% of the original teacher-student gap on the three benchmarks.

Thesis Organization

The remainder of this thesis is structured as follows. Chapter 2 provides background on VLMs, knowledge distillation, distribution matching across mismatched tokenizers, sparse routing and gradient conflict resolution. Chapter 3 presents GRACE in full, covering the Byte-Trie Wasserstein objective, sparse teacher router, gradient-agreement refinement and combined training objective. Chapter 4 describes the experimental setup, datasets and evaluation metrics, baselines and ablation settings. Chapter 5 reports the main comparison, ablation study, efficiency analysis and discussion. A concluding chapter summarizes the main findings, identifies three limitations of the current evaluation and outlines four future directions: repeated-run evaluation, reduced teacher-logit storage and I/O cost, visual token alignment and reasoning-oriented distillation.

Chapter 2

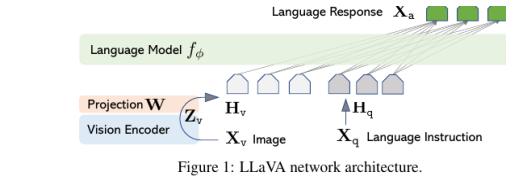
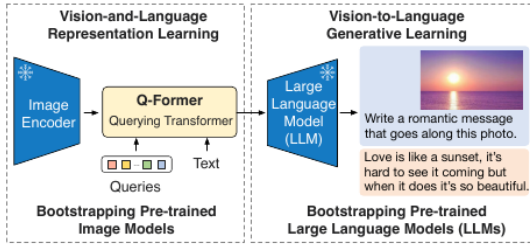
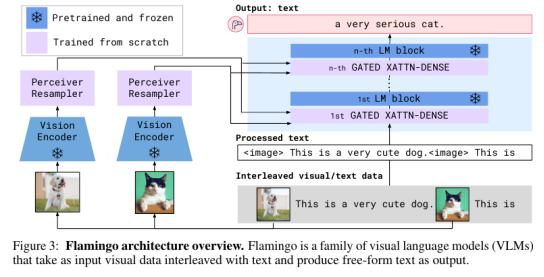
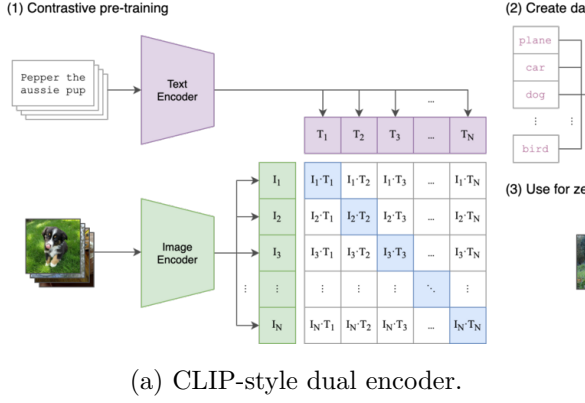
Background and Related Work

Three bodies of prior work bear directly on GRACE: knowledge distillation, tokenizer-agnostic distribution matching and sparse expert routing. The final section shows where these converge into the research gap the thesis addresses.

2.1 Vision-Language Models

Modern vision-language models often combine a vision encoder with a language model. On the visual side, many systems use a Vision Transformer pretrained with an image-text objective, commonly CLIP [31] or SigLIP [32], which converts an image into a sequence of patch embeddings that capture both local and global visual features. A connector module then fuses these visual tokens with text, using cross-attention, a Q-Former or a learned projector [1–3]. This design separates visual perception from language generation: the vision encoder produces a compact sequence of image features, while the language model provides the autoregressive prior used to answer questions. VLM families differ in where and how vision-language fusion happens, as shown in Figure 2.1. Dual-encoder models align image and text embeddings contrastively, cross-attention models let language tokens attend to visual features directly and projector-based models map visual embeddings into the language model hidden space and then decode with a standard causal language model. This thesis focuses on projector-based generative VLMs because their autoregressive output distributions provide the supervision target for distillation. In this setting, projected image embeddings are inserted into the language-model context as visual tokens, so answer generation remains ordinary next-token prediction conditioned on both the image representation and the textual question.

²Panels adapted from CLIP [31], Flamingo [1], BLIP-2 [2] and LLaVA [3].



(c) BLIP-2 with Q-Former.

(d) LLaVA projector-based design.

Figure 2.1: Representative architecture families of modern vision-language models.²

The student VLM produces answer tokens autoregressively. For language modeling, the Transformer maps each hidden state $\mathbf{h}_j$ to next-token logits and probabilities [33]:

$$\mathbf{z}_j = \mathbf{W}_o \mathbf{h}_j + \mathbf{b}_o, \quad p_j(v) = \text{softmax}(\mathbf{z}_j)_v = \frac{\exp(z_{j,v})}{\sum_{u \in \mathcal{V}} \exp(z_{j,u})}. \quad (2.1)$$

At answer position j , the VLM can be abstracted as a mapping from the multimodal input and previous answer tokens to next-token logits:

$$\mathbf{z}_j = f_\theta(x, y_{<j}), \quad \mathbf{z}_j \in \mathbb{R}^{|\mathcal{V}|},$$

where x is a multimodal input and $\mathbf{z}_j$ denotes the logits used for autoregressive answer generation at position j . For an answer sequence $y = (y_1, \dots, y_L)$, the model factorizes the conditional probability as

$$p_\theta(y | x) = \prod_{j=1}^L p_\theta(y_j | x, y_{<j}). \quad (2.2)$$

Token-level distillation therefore compares teacher and student answer-token distributions, $p_{t,j}$ and $p_{s,j}$, at each supervised position [16, 33]. In VQA, short answers such as numbers, names and document fields can differ by a single token, so per-position supervision accuracy matters. Standard KD compares these distributions directly; heterogeneous VLM distillation instead requires a vocabulary-independent matching objective.

2.2 Knowledge Distillation

Knowledge distillation trains a compact student to match the outputs of a larger teacher [16, 17]. Let $\mathcal{L}_{CE}$ denote the supervised cross-entropy loss of the student. A generic distillation objective takes the form

$$\mathcal{L} = \lambda_{CE}\mathcal{L}_{CE} + \lambda_{KD}\mathcal{L}_{KD}, \quad (2.3)$$

where $\mathcal{L}_{KD}$ is a teacher-guided loss. Token-level distillation compares teacher and student output distributions alongside hard labels. The student therefore learns the teacher’s relative probabilities over alternative answers, often called *dark knowledge*, which hard labels discard [16]. For a single answer position, a common same-vocabulary form is

$$\mathcal{L}_{KD} = \tau^2 \text{KL}\left(\mathbf{p}_t^{(\tau)} \parallel \mathbf{p}_s^{(\tau)}\right), \quad (2.4)$$

where τ is the distillation temperature. The cross-entropy term anchors the student to the ground-truth answer, while the KD term transfers the teacher’s relative preference over alternative tokens. The same structure applies in the heterogeneous setting, but the comparison term must handle different teacher and student tokenizers.

The output distributions used in KD are often softened before comparison. Let $\mathbf{z}_s$ and $\mathbf{z}_t$ denote student and teacher logits. With temperatures τ_s and τ_t , the corresponding distributions are

$$\mathbf{p}_s = \text{softmax}\left(\frac{\mathbf{z}_s}{\tau_s}\right), \quad \mathbf{p}_t = \text{softmax}\left(\frac{\mathbf{z}_t}{\tau_t}\right). \quad (2.5)$$

Many classical KD methods compare $\mathbf{p}_s$ and $\mathbf{p}_t$ using divergences such as forward KL or Jensen-Shannon divergence. However, these approaches usually assume that teacher and student share the same vocabulary [16, 21, 34]. The temperature controls how much probability mass is exposed beyond the top answer token. When $\tau > 1$, the softmax distribution becomes flatter and low-rank alternatives receive non-negligible probability. When $\tau < 1$, the distribution becomes sharper and approaches hard-label supervision. The teacher’s tail distribution often encodes semantic alternatives such as related words, number formats or subword variants. The same softening applies before Byte-Trie Wasserstein matching. Since student and teacher temperatures may be chosen independently, τ_t controls how much dark knowledge the teacher exposes, whereas τ_s controls the smoothness of the student distribution under matching.

GRACE restricts distillation to the supervised answer positions, matching the span evaluated at test time [3]. The KD loss therefore masks ignored tokens and truncates

student and teacher answer spans to a common length, producing a signal that works across different teacher and student tokenizers. Let Ω_b^{CE} denote the non-ignored student answer positions for sample b , so the supervised loss is

$$\mathcal{L}_{CE}^{(b)} = \frac{1}{|\Omega_b^{CE}|} \sum_{j \in \Omega_b^{CE}} -\log p_s(y_j \mid x_b, y_{<j}). \quad (2.6)$$

The same masking applies to KD positions, keeping the teacher signal on the evaluated answer span rather than on prompt reconstruction, padding or formatting. For multiple teachers, the KD position set can be teacher-specific because teacher tokenizers and generated answer spans may differ. Per-teacher KD positions $\Omega_{b,i}^{KD}$ are constructed separately for each teacher, restricted to valid aligned answer positions.

KD methods differ in the type of teacher signal they use. Common response-level divergences and feature-level similarity losses are collected in Table 2.1 [16, 34]. In the table, p and q denote teacher and student output distributions over token events t , and $m = \frac{1}{2}(p + q)$ is the JS midpoint distribution. The teacher and student feature vectors are

$$\mathbf{u} = \Phi_T(f_T(x, y)), \quad \mathbf{v} = \Phi_S(f_S(x, y)).$$

The index r marks feature dimensions or normalized feature bins. *Response-based* distillation uses the teacher’s output distribution as supervision. In its standard form, the loss compares next-token distributions:

$$\mathcal{L}_{resp} = \frac{1}{|\Omega|} \sum_{j \in \Omega} \text{KL}(\mathbf{p}_{t,j}^{(\tau)} \parallel \mathbf{p}_{s,j}^{(\tau)}), \quad (2.7)$$

where Ω is the set of supervised answer positions. This is the classical KD form, introduced by Hinton et al. [16] and still common in language-model and VLM compression [17, 35].

Feature-based distillation aligns intermediate representations between student and teacher layers. Given student and teacher hidden states $\mathbf{H}_s^{(\ell)}$ and $\mathbf{H}_t^{(m)}$ at selected layers, a direct matching objective is

$$\mathcal{L}_{feat} = \left\| \phi_s(\mathbf{H}_s^{(\ell)}) - \phi_t(\mathbf{H}_t^{(m)}) \right\|_2^2, \quad (2.8)$$

where ϕ_s and ϕ_t are projection functions that make the hidden states comparable. Feature-based methods apply when output vocabularies are incompatible, since the comparison is defined in representation space without direct vocabulary comparison [24, 36]. However, they predict answer likelihood less directly and can transfer internal structure specific to the teacher architecture, so this thesis treats response-level and representation-level signals as complementary [36, 37].

Table 2.1: Summary of response-level and feature-level KD objectives.

Response-Level Divergences		Feature-Level Losses	
Type	$D(p, q)$	Function $\mathcal{L}_F$	Expression
Forward KLD	$\sum_t p(t) \log \frac{p(t)}{q(t)}$	L2-Norm Distance	$\ \mathbf{u} - \mathbf{v}\ _2$
Reverse KLD	$\sum_t q(t) \log \frac{q(t)}{p(t)}$	L1-Norm Distance	$\ \mathbf{u} - \mathbf{v}\ _1$
JS Divergence	$\frac{1}{2} \text{KL}(p\ m) + \frac{1}{2} \text{KL}(q\ m)$	Cross-Entropy Loss	$-\sum_r \mathbf{u}_r \log \mathbf{v}_r$

2.3 Tokenizer-Agnostic Distribution Matching

Same-vocabulary KD assumes that teacher and student probabilities are indexed by the same token set. Heterogeneous VLM distillation breaks this assumption: a teacher may tokenize the string “invoice” as one token while the student splits it into several subword pieces; directly applying KL over their respective vocabularies then compares unrelated indices and is invalid [21]. The natural fix is to decode each token into a string and compare probability mass over decoded strings instead of vocabulary indices. Let $\mathcal{V}_t$ and $\mathcal{V}_s$ be the teacher and student vocabularies with decoding maps into a shared string space $\mathcal{S}$:

$$\psi_t : \mathcal{V}_t \rightarrow \mathcal{S}, \quad \psi_s : \mathcal{V}_s \rightarrow \mathcal{S}.$$

Transport-based objectives replace same-vocabulary KL with a distance over probability mass on a shared string support.

Optimal transport compares two probability measures by the minimum cost required to move mass from one to the other [38]. For probability measures μ_s and μ_t , the p -Wasserstein cost is

$$W_p^p(\mu_s, \mu_t) = \min_{\gamma \in \Pi(\mu_s, \mu_t)} \int_{\Omega_s \times \Omega_t} \|x - y\|^p d\gamma(x, y), \quad (2.9)$$

where $\Pi(\mu_s, \mu_t)$ is the set of joint couplings with marginals μ_s and μ_t . For discrete weights $\mathbf{a}$, $\mathbf{b}$ and cost matrix $\mathbf{C}$, this becomes

$$W(\mathbf{a}, \mathbf{b}) = \min_{\mathbf{P} \in \mathcal{U}(\mathbf{a}, \mathbf{b})} \langle \mathbf{P}, \mathbf{C} \rangle, \quad \mathcal{U}(\mathbf{a}, \mathbf{b}) = \{\mathbf{P} \geq 0 \mid \mathbf{P}\mathbf{1} = \mathbf{a}, \mathbf{P}^\top \mathbf{1} = \mathbf{b}\}. \quad (2.10)$$

Unlike pointwise divergences, Wasserstein distances reflect displacement even when supports are disjoint, as Figure 2.2 illustrates.

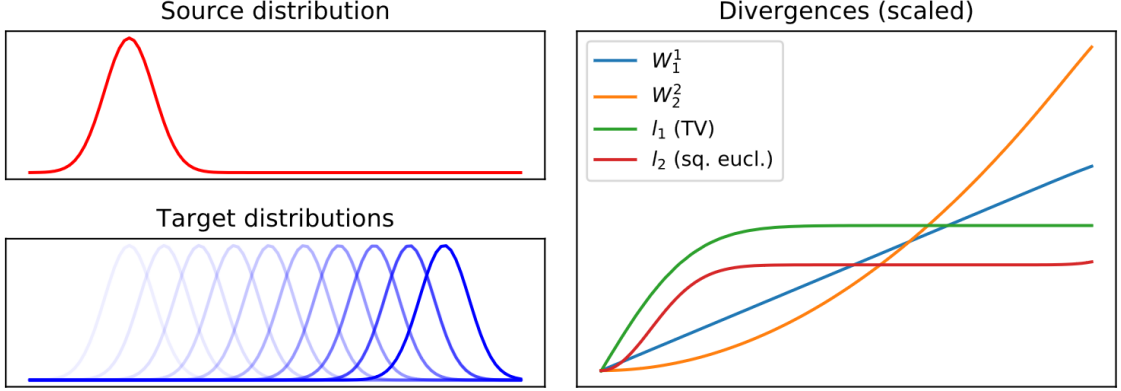


Figure 2.2: Geometric behavior of Wasserstein distance.¹

For tokenizer-agnostic matching, dense OT is still costly because it requires a pairwise ground cost between all source and target bins. A *tree metric* replaces the pairwise cost matrix: probability mass is placed on a shared hierarchy and the distance reduces to the sum of weighted *subtree-mass imbalances* [22, 38]. For a rooted tree $T = (V, E)$ with edge weights $w(e)$, one convenient form is

$$W_T(\mu, \nu) = \sum_{e \in E} w(e) |\mu(\Gamma(e)) - \nu(\Gamma(e))|, \quad (2.11)$$

where $\Gamma(e)$ denotes the subtree below edge e . On a tree metric, transport reduces to per-edge subtree-mass imbalances, avoiding the dense source-target coupling of standard OT [22, 23]. Figures 2.3a and 2.3b show the two commonly used OT views: a tree-metric view for fast subtree-mass comparisons and a matrix view for discrete transport plans. Tree-sliced OT shows that averaging over random tree metrics preserves OT geometry while admitting fast closed-form computation [23].

A different approach avoids output distributions by matching intermediate representations directly. Let the student and teacher hidden-state matrices be

$$\mathbf{H}_s \in \mathbb{R}^{n_s \times d_s}, \quad \mathbf{H}_t \in \mathbb{R}^{n_t \times d_t}$$

Because the sequence lengths and dimensions may differ, alignment usually requires pooling, projection, interpolation or layer selection:

$$\tilde{\mathbf{H}}_s = A_s(\mathbf{H}_s), \quad \tilde{\mathbf{H}}_t = A_t(\mathbf{H}_t), \quad \mathcal{L}_{hid} = \left\| \tilde{\mathbf{H}}_s - \tilde{\mathbf{H}}_t \right\|_F^2. \quad (2.12)$$

Here, A_s and A_t denote alignment operations that place student and teacher activations in a comparable space. This family applies to heterogeneous models because it

¹Adapted from Flamary and Courty [39].

¹Panel (a) is adapted from Takezawa et al. [22]. Panel (b) is adapted from Wikimedia Commons and Peyré and Cuturi [38, 40].

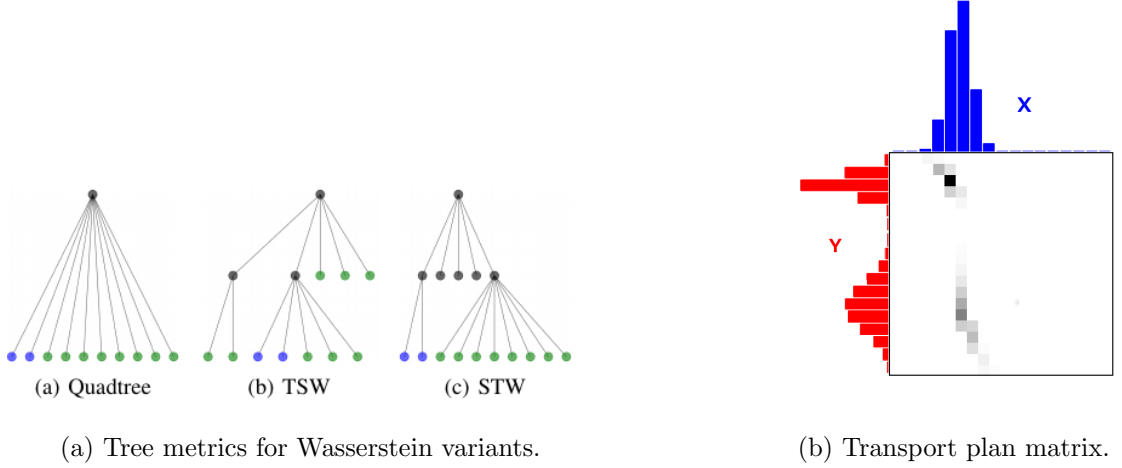


Figure 2.3: Discrete optimal transport views.¹

avoids direct vocabulary comparison and can draw on visual, textual or fused multi-modal representations [36].

Centered Kernel Alignment (CKA) compares representations through their similarity structure and avoids direct elementwise hidden-state matching [24]. Given activation matrices $\mathbf{X} \in \mathbb{R}^{n \times d_x}$ and $\mathbf{Y} \in \mathbb{R}^{n \times d_y}$ over the same n examples, the linear CKA score is

$$\text{CKA}(\mathbf{X}, \mathbf{Y}) = \frac{\|\mathbf{Y}^\top \mathbf{X}\|_F^2}{\|\mathbf{X}^\top \mathbf{X}\|_F \|\mathbf{Y}^\top \mathbf{Y}\|_F}. \quad (2.13)$$

Higher CKA values mean the two representations share the same pairwise similarity structure across the evaluated samples. Because CKA is invariant to isotropic scaling and orthogonal transformations, it remains valid when teacher and student have different hidden dimensions, unlike direct hidden-state matching. This thesis uses CKA both as a distillation signal and as a diagnostic for teacher-student feature alignment [24, 37].

2.4 Sparse Routing and Expert Selection

Recent sparse expert models route each sample or token through only a subset of available experts, reducing computation to a fraction of parameters per input [41–43]. Let $\mathbf{h}^{(b)}$ denote a pooled representation for sample b . A router network $G_\phi(\cdot)$ produces gate logits

$$\mathbf{r}^{(b)} = G_\phi(\mathbf{h}^{(b)}), \quad \tilde{\mathbf{r}}^{(b)} = \text{AdjustedScores}(\mathbf{r}^{(b)}), \quad \mathbf{p}^{(b)} = \text{softmax}(\tilde{\mathbf{r}}^{(b)}), \quad (2.14)$$

where $p_i^{(b)}$ is the initial routing weight for expert i . The adjusted scores $\tilde{\mathbf{r}}^{(b)}$ may include expert bias, training noise or temperature scaling before the routing softmax. *Top- k routing* retains only the highest-scoring experts and zeros out all others, while *capacity constraints* prevent any single expert from receiving too many assignments. The selected expert set is

$$\mathcal{S}_b = \text{TopK}(\mathbf{p}^{(b)}, k).$$

The sparse expert output can be written as

$$\mathbf{y}^{(b)} = \sum_{i \in \mathcal{S}_b} \alpha_i^{(b)} E_i(\mathbf{h}^{(b)}), \quad \alpha_i^{(b)} = \frac{p_i^{(b)}}{\sum_{j \in \mathcal{S}_b} p_j^{(b)}}. \quad (2.15)$$

Only the selected experts are evaluated, keeping active computation small even when the total expert count is large. The same abstraction transfers directly to teacher routing: each teacher acts as a frozen expert that contributes a weighted distillation loss for a subset of samples.

Sparse routing also needs mechanisms that prevent collapse onto a small subset of experts. In a sparse MoE layer, collapse is both a modeling problem and a systems problem: the overloaded experts determine the step latency, while under-used experts receive few gradients and remain poorly specialized [41, 44, 45]. Let a mini-batch contain T routed items and let N denote the number of experts. For item b , the router produces probabilities $\mathbf{p}^{(b)} \in \Delta^{N-1}$. A top- k router induces the binary assignment mask

$$\mathbf{a}_i^{(b)} = \mathbf{1}\{i \in \text{TopK}(\mathbf{p}^{(b)}, k)\}, \quad n_i = \sum_{b=1}^T \mathbf{a}_i^{(b)}, \quad (2.16)$$

where n_i is the number of routed items assigned to expert i . Balanced routing corresponds to

$$n_i \approx \frac{kT}{N}.$$

When n_i is much larger than this value for a few experts, the router has effectively reduced the model to a small ensemble of busy experts while wasting the capacity of the remaining ones. The routing statistics used by many auxiliary losses can be written as a *soft* load and a *hard* load:

$$\bar{p}_i = \frac{1}{T} \sum_{b=1}^T p_i^{(b)}, \quad \bar{q}_i = \frac{1}{kT} \sum_{b=1}^T \mathbf{a}_i^{(b)}. \quad (2.17)$$

Here, $\bar{p}_i$ measures how much probability mass the router gives to expert i before hard selection, whereas $\bar{q}_i$ measures how often the expert is actually selected after sparse routing. The Switch Transformer balancing loss uses their scaled dot product [42]:

$$\mathcal{L}_{bal} = \alpha_{bal} N \sum_{i=1}^N \bar{p}_i \bar{q}_i. \quad (2.18)$$

The factor N keeps the magnitude approximately independent of the number of experts because the uniform solution gives

$$\sum_i (1/N)(1/N) = \frac{1}{N}.$$

The hard load $\bar{q}_i$ is non-differentiable through the top- k decision, but $\bar{p}_i$ is differentiable. The loss therefore gives the router a gradient signal that discourages assigning high average probability to already busy experts.

In GRACE, entropy and z -loss terms stabilize sparse teacher selection while the task and distillation losses remain the main training objectives.

Sparse routing is most effective when the best expert for a given input depends on that input [41, 42]. In a generic sample-adaptive router, the selected expert set is a function of a sample representation:

$$\mathcal{S}_b = \text{TopK}(\text{softmax}(G_\phi(\mathbf{h}^{(b)})), k). \quad (2.19)$$

The loss is then computed only from experts in $\mathcal{S}_b$. This differs from fixed ensembling, where every expert contributes to every example with the same global weight. In multimodal QA, one sample may require OCR and document layout reasoning, another chart interpretation, and a third natural image recognition.

A global weighting scheme uses one vector $\boldsymbol{\pi} \in \Delta^{N-1}$ for the full training set:

$$\mathcal{L}^{(b)} = \sum_{i=1}^N \pi_i \mathcal{L}_i^{(b)}. \quad (2.20)$$

A sample-adaptive scheme uses $w_i^{(b)}$:

$$\mathcal{L}^{(b)} = \sum_{i=1}^N w_i^{(b)} \mathcal{L}_i^{(b)}, \quad w_i^{(b)} = 0 \text{ for } i \notin \mathcal{S}_b. \quad (2.21)$$

The sparsity constraint caps the number of active experts and limits the influence of low-utility teachers on any given update.

Distillation can reuse this routing framework by treating frozen teachers as experts, with the router selecting which teachers supervise each sample. Teacher T_i contributes loss $\mathcal{L}_{KD}^{(b,i)}$ for sample b . A routed distillation loss takes the form

$$\mathcal{L}_{route}^{(b)} = \sum_{i \in \mathcal{S}_b} \omega_i^{(b)} \mathcal{L}_{KD}^{(b,i)}. \quad (2.22)$$

Teacher routing is then MoE selection applied to distillation: it determines which teacher losses update the student for each sample. Unlike standard MoE, the experts here are frozen, so only the student and router parameters receive gradient updates. Routing in a distillation setting faces an additional complication: teacher quality varies with the input, the current student state and the task objective. A high router score can still produce a KD gradient that conflicts with supervised learning, so GRACE adds gradient agreement after routing [25, 46, 47].

2.5 Multi-Teacher Distillation

Multi-teacher distillation uses several teachers rather than one. Let $\mathcal{T} = \{T_1, \dots, T_N\}$ be the teacher pool, and let $\mathcal{L}_{KD}^{(b,i)}$ denote the distillation loss produced by teacher T_i for sample b . The simplest aggregation rule is uniform averaging:

$$\mathcal{L}_{KD}^{(b)} = \frac{1}{N} \sum_{i=1}^N \mathcal{L}_{KD}^{(b,i)}. \quad (2.23)$$

Uniform averaging is useful as a reference because it isolates teacher diversity from any learned selection. Its limitation is that it assigns equal influence to all teachers, even when their strengths vary across document images, charts and natural scene text.

A heterogeneous teacher pool can differ along multiple axes: tokenizer, language backbone, vision encoder, pretraining data and answer calibration. One model may provide stronger supervision for OCR-heavy samples while another is better calibrated on charts. But teacher distributions can also disagree on the same answer position when their tokenizers split the answer string differently. A multi-teacher method must therefore select and weight teachers, not just aggregate them.

A more general multi-teacher objective assigns weights to teachers:

$$\mathcal{L}_{KD}^{(b)} = \sum_{i=1}^N w_i^{(b)} \mathcal{L}_{KD}^{(b,i)}, \quad w_i^{(b)} \geq 0, \quad \sum_{i=1}^N w_i^{(b)} = 1. \quad (2.24)$$

The weights can be fixed globally, learned globally or conditioned on the sample. Fixed weights ignore input variation entirely. Globally learned weights can identify a single dominant teacher but give every sample the same mixture. VLM tasks require sample-dependent weights: the best teacher for scene-text OCR may differ from the best for chart reading or document layout. Existing adaptive multi-teacher methods that address this setting are surveyed in Section 2.6.

Sparse routing implements sample-dependent weighting directly. The routing framework in Section 2.4 can be applied to teacher selection by treating frozen teachers as experts whose losses are selected for each sample. A routed teacher set reduces the teacher losses per update and makes teacher-sample assignments explicit. However, routing introduces two design questions that standard ensembling does not address. First, the router must avoid collapsing onto a single teacher too early, otherwise the teacher pool becomes an expensive single-teacher system. Second, a teacher that appears relevant to the input may still produce a KD direction that conflicts with the supervised CE objective.

The second issue is an optimization conflict rather than only a selection problem. Consider

the standard combined distillation loss

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha \mathcal{L}_{KD}.$$

Because both terms share the same student parameters, the combined update reinforces learning when CE and KD gradients agree and partially offsets learning when they point in competing directions. This is the multi-objective optimization setting studied by gradient-conflict methods [46, 47]. In multi-teacher distillation, each teacher contributes a separate KD gradient, so selecting a teacher subset does not by itself verify that the selected gradients support the supervised update. GRACE therefore uses routing to propose candidate teachers and gradient-agreement refinement to check whether their KD gradients align with the supervised CE objective.

2.6 Related Work

The thesis builds on four lines of prior work: response-based KD, cross-tokenizer distribution matching, representation matching, and adaptive multi-teacher selection. Each addresses part of the problem, but no prior method handles all three constraints together.

Response-based and vocabulary-mismatched KD. Response-based KD transfers teacher output distributions to the student [16, 17]. In multimodal models, it has been used for compressing VLMs or transferring vision-language behavior from larger models to smaller ones [35, 48, 49]. This family supervises the output distribution used at prediction time, which directly matches the generation objective. Its main limitation in heterogeneous teacher-student settings is the shared-vocabulary assumption: KL, reverse KL and Jensen-Shannon losses compare coordinates that are only meaningful when teacher and student token indices refer to the same vocabulary items.

Cross-tokenizer KD methods relax this assumption by comparing decoded strings or probability mass over a shared support [21, 36]. Optimal-transport and tree-Wasserstein methods are especially relevant because they define a ground geometry between tokens rather than requiring exact token-index equality [22, 23]. These methods provide the vocabulary-independent loss that GRACE uses at the output level. All are formulated as single-teacher objectives, however: none decides which teacher should supervise a given sample, and none checks whether a chosen teacher’s KD gradient conflicts with the supervised CE objective.

Representation matching and CKA alignment. Representation-level distillation avoids the vocabulary problem by matching hidden states, attention maps or similarity structure rather than output-token probabilities. CKA is a common tool for comparing

representations because it measures similarity structure while being invariant to orthogonal changes of basis and isotropic scaling [24]. This makes CKA applicable even when teacher and student tokenizers or hidden dimensions differ. Recent cross-tokenizer work also uses representation-level information as a complementary alignment signal [36].

The limitation is that representation similarity is not the same as answer distribution matching. Two models can have similar hidden-state geometry while their language heads assign different probabilities to answer tokens. This gap matters directly for VQA, where a small probability shift among OCR strings or numeric answers changes the final evaluated score. Representation matching also leaves the teacher-selection question unresolved: it can compare a teacher and student once a teacher has been chosen, but it does not specify which teacher should dominate for a given sample. For these reasons, this thesis treats representation matching as a diagnostic baseline and keeps output-level distribution matching as the primary distillation signal.

Multi-teacher KD and sparse routing. Multi-teacher KD studies how to combine several teachers rather than relying on a single supervisor. Reinforcement-based and meta-learning approaches learn teacher weights or selection policies that adapt over training [25–27]. These methods show that teacher diversity raises student accuracy when teachers make complementary errors. They also identify a core challenge: teacher choice must be sample-dependent, because the best teacher for document layout may differ from the best teacher for chart reasoning or natural scene text.

Sparse expert routing selects only a subset of experts per input [41, 42, 45]. In a distillation setting, the experts are frozen teachers and the selected objects are teacher losses rather than forward-pass expert activations. The distinction is that routing only estimates teacher relevance. A routed teacher can still produce a KD gradient that conflicts with the supervised CE objective, so sparse routing must be paired with an objective-compatibility check.

Gradient-conflict resolution. Gradient-conflict methods measure whether multiple objectives induce compatible updates on shared parameters. Student-customized distillation and gradient surgery methods show that conflicting gradients can harm learning and that gradient direction filters or reweights auxiliary objectives [46, 47]. These ideas apply directly to multi-teacher KD, where each teacher contributes a separate KD gradient. However, prior gradient-conflict methods do not address cross-tokenizer output matching, and they do not provide a sparse teacher-selection mechanism for a heterogeneous VLM teacher pool.

No existing method closes all four gaps simultaneously. Vocabulary-aligned KD breaks across tokenizers. Cross-tokenizer transport fixes that but assumes a single teacher. Representation matching avoids the vocabulary problem but produces no answer-calibrated

signal and leaves teacher selection unresolved. Routing-based methods select teachers but none verify gradient compatibility. GRACE addresses all four within one framework by combining a Byte-Trie Wasserstein output loss, sparse teacher routing and gradient-agreement refinement.

Chapter 3

Proposed Method

GRACE addresses three constraints that existing methods handle separately: vocabulary mismatch between teacher and student, variable teacher usefulness across samples, and KD gradients that conflict with the supervised CE objective. Each component of the framework resolves the limitation the previous one leaves open.

3.1 Method Overview and Problem Formulation

We consider a training set

$$\mathcal{D} = \{(x_n, y_n)\}_{n=1}^{N_{\text{data}}},$$

where x_n is a multimodal input and y_n is the target answer sequence.

Let S_θ denote the student VLM with trainable parameters θ , let G_ϕ denote the trainable teacher router with parameters ϕ and let $\mathcal{T} = \{T_i\}_{i=1}^N$ be the frozen teacher pool. For each sample b , the student produces logits $\mathbf{z}_s^{(b)}$ and teacher T_i provides logits $\mathbf{z}_t^{(b,i)}$.

Three constraints shape the GRACE design. First, teacher and student output spaces use different token indices, so the KD loss compares decoded token content through a shared string space. Second, teacher usefulness is sample-dependent: a teacher that is useful for document layout may be less useful for chart reasoning or natural scene text. Third, teacher relevance and optimization compatibility can diverge when a teacher’s logits look correct but its gradient conflicts with the CE objective. GRACE addresses these constraints with Byte-Trie Wasserstein matching, sparse teacher routing and gradient-agreement refinement.

Figure 3.1 summarizes the full training pipeline. The router uses pooled student hidden states to select a sparse teacher subset; Byte-Trie Wasserstein KD then computes a per-teacher loss by matching decoded token paths through a shared byte trie. Gradient-agreement refinement adjusts the selected teacher weights by how well each teacher’s KD gradient aligns with the supervised CE objective. The final objective combines supervised cross-entropy, this routed KD signal and two router regularization terms [41, 42, 46].

A GRACE update starts from a mini-batch

$$\mathcal{B} = \{(x_b, y_b)\}_{b=1}^B$$

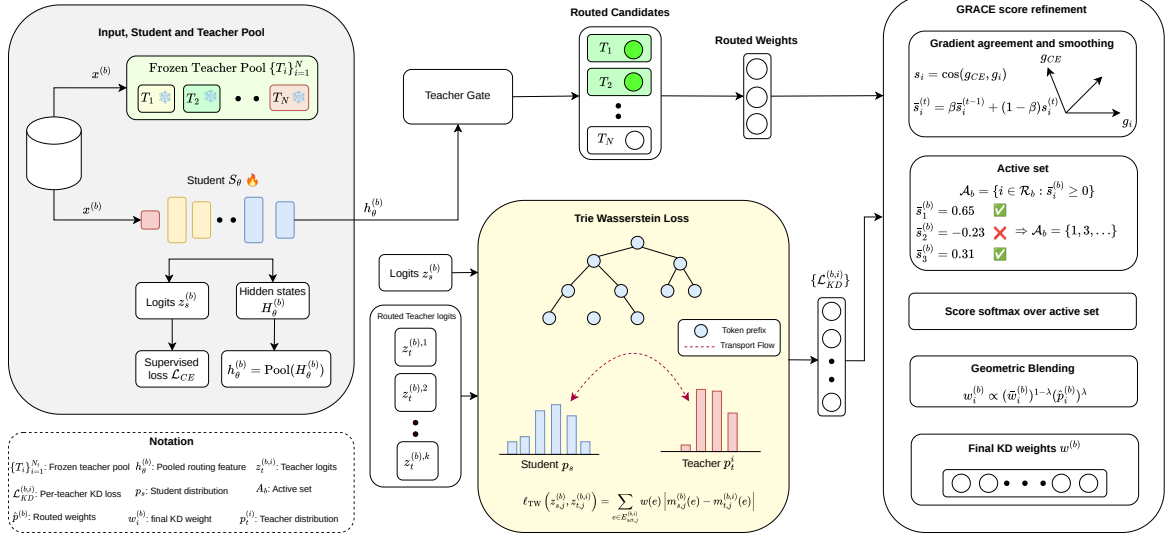


Figure 3.1: Overview of the GRACE architecture.

and the frozen teacher pool $\mathcal{T}$. The student first produces logits, hidden states and the supervised cross-entropy loss for the answer tokens. The router then uses the pooled student hidden state to select a sparse routed set $\mathcal{R}_b$ for each sample. For every selected teacher, Byte-Trie Wasserstein KD computes a per-teacher loss $\mathcal{L}_{KD}^{(b,i)}$, using the shared byte-trie geometry to compare student and teacher output distributions. Gradient-agreement refinement then adjusts routed teacher weights by comparing each teacher’s KD gradient with the supervised CE gradient. The routed KD signal for sample b is the weighted combination of teacher-specific losses:

$$\bar{\mathcal{L}}_{KD}^{(b)} = \sum_{i \in \mathcal{R}_b} w_i^{(b)} \mathcal{L}_{KD}^{(b,i)}.$$

The final weights $w_i^{(b)}$ are determined by both the router and the gradient-agreement refinement step. The method introduces several hyperparameters (ρ_{trie} , η_{boundary} , K_{TW} , k_{route} , β_{EMA} , λ_{blend} and others); their values and the ablation study that isolates the most influential ones are reported in Appendix B and Section 5.2 respectively. The main notation used throughout the method is listed in Table 3.1.

Table 3.1: Notation table.

Symbol	Meaning
$\mathcal{D}, \mathcal{B}$	Training set and mini-batch.
$S_\theta, \mathcal{T} = \{T_i\}_{i=1}^N, G_\phi$	Trainable student VLM, frozen teacher pool and teacher routing network.
$\mathbf{z}_s^{(b)}, \mathbf{z}_t^{(b,i)}, p_{s,j}^{(b)}, p_{t,j}^{(b,i)}$	Student and teacher logits and softened distributions.
$\Omega_b^{CE}, \Omega_{b,i}^{KD}$	Supervised CE positions and matched KD positions.

Symbol	Meaning
$\mathbf{H}_s, \mathbf{h}$	Student hidden states and pooled routing feature.
$\rho_{\text{trie}}, \eta_{\text{boundary}}, K_{\text{TW}}, e_{\text{tail}}$	Byte-trie edge decay, boundary-edge weight, sparse support size and residual tail edge.
$\mathcal{K}_{s,j}^{(b)}, \mathcal{K}_{t,j}^{(b,i)}, r_{s,j}^{(b)}, r_{t,j}^{(b,i)}$	Retained top- K_{TW} token sets and residual tail masses.
$m_{s,j}^{(b)}(e), m_{t,j}^{(b,i)}(e), \ell_{\text{TW}}$	Sparse subtree masses and token-level Byte-Trie Wasserstein loss.
$k_{\text{route}}, \mathbf{r}, \mathbf{p}, \mathcal{R}_b, \hat{\mathbf{p}}^{(b)}$	Router top- k , router logits, soft router weights, routed teacher set and renormalized routed weights.
$s_i, \hat{s}_i, \mathcal{A}_b, \boldsymbol{\omega}^{(b)}, \mathbf{w}^{(b)}$	Agreement scores, active routed set, score-softmax weights and final KD weights.
$\mathbf{g}_{\text{CE}}, \mathbf{g}_i, \delta, \beta_{\text{EMA}}, \beta_{\text{score}}$	CE and KD gradients, agreement threshold, EMA decay and score-softmax sharpness.
$\mathcal{L}_{\text{CE}}, \mathcal{L}_{\text{KD}}^{(b,i)}, \bar{\mathcal{L}}_{\text{KD}}^{(b)}, \mathcal{L}$	Supervised CE loss, per-teacher KD loss, sample-level routed KD loss and final objective.
$\mathcal{L}_{\text{ent}}, \mathcal{L}_z, \alpha_{\text{KD}}, \lambda_{\text{ent}}, \lambda_z$	Router regularizers and final-objective weights.

3.2 Byte-Trie Wasserstein Distillation

For sample b and teacher T_i , GRACE restricts KD to matched answer positions $\Omega_{b,i}^{KD}$, after masking ignored labels and truncating student and teacher answer spans to a common length. At each position $j \in \Omega_{b,i}^{KD}$, the student and teacher logits define softened distributions

$$p_{s,j}^{(b)} = \text{softmax}\left(z_{s,j}^{(b)} / \tau_s\right), \quad p_{t,j}^{(b,i)} = \text{softmax}\left(z_{t,j}^{(b,i)} / \tau_t\right).$$

The remaining computation has three steps: map decoded tokens into a shared byte trie, approximate the full vocabulary distributions with sparse retained supports and residual tail mass, and compute the tree-Wasserstein loss from subtree-mass imbalances.

3.2.1 Shared Byte Trie

Applying KD by token index requires aligned vocabularies, but with different tokenizers the same index may correspond to unrelated decoded strings. GRACE therefore maps decoded teacher and student tokens into a shared UTF-8 byte trie. Each token’s boundary markers are separated from its lexical content; the content is normalized and encoded as bytes, then terminated with an EOS sentinel so that complete tokens remain distinct from strict prefixes.

This canonicalization suppresses tokenizer artifacts during trie-based matching. Many tokenizers encode word boundaries with visible markers or leading whitespace, even when the underlying answer string is the same. GRACE separates this boundary information from the byte path of the token content. The normalized content determines the main trie path, while boundary markers are represented by optional low-cost boundary edges. Tokens sharing lexical content therefore land close in the trie, while the low-weight boundary edges still penalize whitespace differences when present.

A document-style answer string such as “invoice”, in the DocVQA setting [5], illustrates the problem. A tokenizer with an explicit word-boundary marker may produce it as a single token `<wb>invoice`, where `<wb>` marks the word boundary, while another tokenizer splits it into `in` and `voice`. Without canonicalization, `<wb>invoice` and `in` share no index and the loss cannot relate them. After canonicalization, the boundary marker is stripped from the main path and placed on a low-cost boundary edge; the byte sequence `invoice` is inserted into the trie, where `in` already occupies the same prefix. Transport between `<wb>invoice` and `in` is therefore low, reflecting the genuine lexical overlap rather than an index coincidence.

Appending an EOS sentinel prevents prefix collapse by creating distinct terminal paths, so `doc` is distinguishable from the prefix of `document`.

The trie defines the ground-cost geometry for token comparison. Content and EOS edges receive weights that decay with depth, while optional boundary edges receive a small fixed weight:

$$w(e) = \begin{cases} \rho_{\text{trie}}^{d(e)-1}, & e \in E_{\text{content}} \cup E_{\text{EOS}}, \\ \eta_{\text{boundary}}, & e \in E_{\text{boundary}}. \end{cases} \quad (3.1)$$

where $0 < \rho_{\text{trie}} < 1$, $\eta_{\text{boundary}} \geq 0$ and $d(e)$ is the edge depth. Thus, tokens sharing long byte prefixes receive low transport cost under the trie metric even if they come from different tokenizers [22, 38].

3.2.2 Sparse Support and Residual Tail Edge

Using the complete vocabulary trie at every answer position is computationally prohibitive, scaling with the number of vocabulary entries per position. GRACE therefore keeps only the top- K_{TW} student and teacher tokens after masking ignored entries. The retained tokens are inserted along their canonical trie paths, while all omitted probability mass is collapsed into a residual tail edge e_{tail} . This preserves total probability mass while keeping the active edge set small.

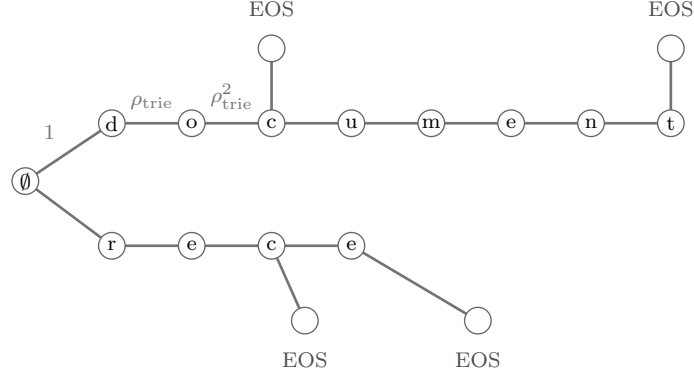


Figure 3.2: Shared byte-trie tokenizer matching.

Let K_{TW} denote the sparse trie support size. Let $\mathcal{V}_s^*$ and $\mathcal{V}_t^*$ denote the non-ignored student and teacher vocabularies. Before constructing the sparse trie, the student and teacher probabilities are renormalized over the non-ignored vocabularies so that the retained and residual masses form valid distributions. Ignored entries are excluded from the softmax normalizer before top- K_{TW} selection. Let $\mathcal{K}_{s,j}^{(b)} \subset \mathcal{V}_s^*$ and $\mathcal{K}_{t,j}^{(b,i)} \subset \mathcal{V}_t^*$ be the retained token sets. The residual mass assigned to the tail edge is

$$r_{s,j}^{(b)} = 1 - \sum_{v \in \mathcal{K}_{s,j}^{(b)}} p_{s,j}^{(b)}(v), \quad r_{t,j}^{(b,i)} = 1 - \sum_{v \in \mathcal{K}_{t,j}^{(b,i)}} p_{t,j}^{(b,i)}(v). \quad (3.2)$$

The residual tail preserves probability mass outside the retained top- K_{TW} support. For the student distribution,

$$\sum_{v \in \mathcal{K}_{s,j}^{(b)}} p_{s,j}^{(b)}(v) + r_{s,j}^{(b)} = 1,$$

and the same identity holds for the teacher. The retained top- K_{TW} paths are compared explicitly while the rest of the probability mass is pooled into the shared tail. When teacher distributions are softened, non-trivial probability mass can lie outside the top answer token.

Let $\pi(v)$ be the trie path for token v , including canonical content edges, the EOS edge and any optional boundary edge. Let e_{tail} be the global residual tail edge with $w(e_{\text{tail}}) = 1$. For any active trie edge e , the sparse subtree masses are

$$m_{s,j}^{(b)}(e) = \sum_{\substack{v \in \mathcal{K}_{s,j}^{(b)} \\ e \in \pi(v)}} p_{s,j}^{(b)}(v) + \mathbf{1}[e = e_{\text{tail}}] r_{s,j}^{(b)}, \quad (3.3)$$

$$m_{t,j}^{(b,i)}(e) = \sum_{\substack{v \in \mathcal{K}_{t,j}^{(b,i)} \\ e \in \pi(v)}} p_{t,j}^{(b,i)}(v) + \mathbf{1}[e = e_{\text{tail}}] r_{t,j}^{(b,i)}. \quad (3.4)$$

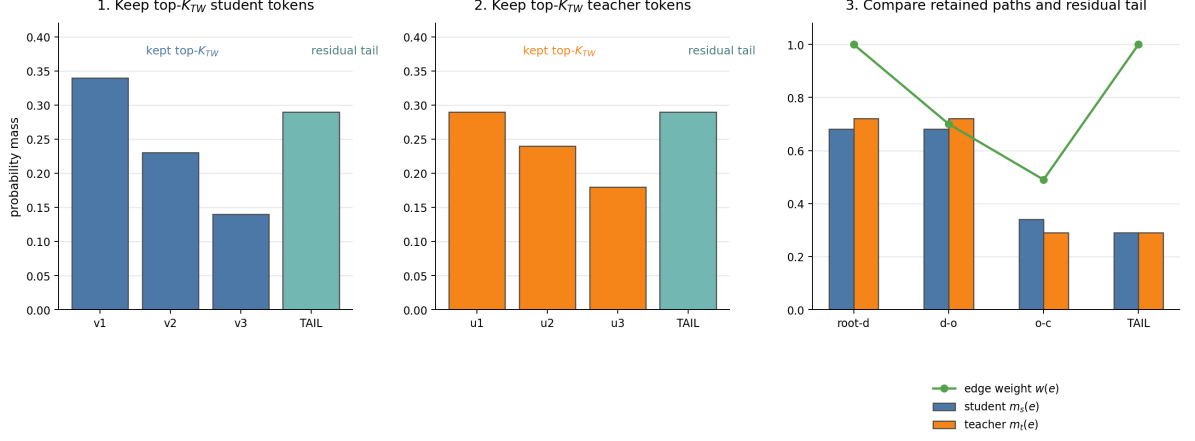


Figure 3.3: Sparse Byte-Trie Wasserstein support.

3.2.3 Byte-Trie Wasserstein Objective

Given the shared trie, each distribution induces subtree masses on active edges. In the dense form, let $\mathcal{V}_s$ and $\mathcal{V}_t$ denote the student and teacher token vocabularies at a given answer position. For edge e , $m_s(e)$ and $m_t(e)$ measure how much student and teacher probability passes through that edge:

$$m_s(e) = \sum_{\substack{v \in \mathcal{V}_s \\ e \in \pi(v)}} p_s(v), \quad m_t(e) = \sum_{\substack{u \in \mathcal{V}_t \\ e \in \pi(u)}} p_t(u).$$

On a tree metric, the Wasserstein distance has the closed form

$$W_T(p_s, p_t) = \sum_{e \in E} w(e) |m_s(e) - m_t(e)|. \quad (3.5)$$

This form follows from the standard W_1 Kantorovich optimal transport problem under the trie-induced tree metric [22, 38]. The full proof is provided in Appendix A. In training, GRACE evaluates the same per-edge formula on the sparse active edge set defined above, using the retained token paths and residual tail edge. The token-level Byte-Trie Wasserstein loss is

$$\ell_{TW}(\mathbf{z}_{s,j}^{(b)}, \mathbf{z}_{t,j}^{(b,i)}) = \sum_{e \in \mathcal{E}_{act,j}^{(b,i)}} w(e) |m_{s,j}^{(b)}(e) - m_{t,j}^{(b,i)}(e)|, \quad (3.6)$$

where $\mathcal{E}_{act,j}^{(b,i)}$ contains the active canonical content edges, EOS edges, optional boundary edges and the residual tail edge from the student and teacher supports.

The per-sample, per-teacher KD loss is the mean over supervised answer tokens:

$$\mathcal{L}_{KD}^{(b,i)} = \frac{1}{|\Omega_{b,i}^{KD}|} \sum_{j \in \Omega_{b,i}^{KD}} \ell_{TW}(\mathbf{z}_{s,j}^{(b)}, \mathbf{z}_{t,j}^{(b,i)}). \quad (3.7)$$

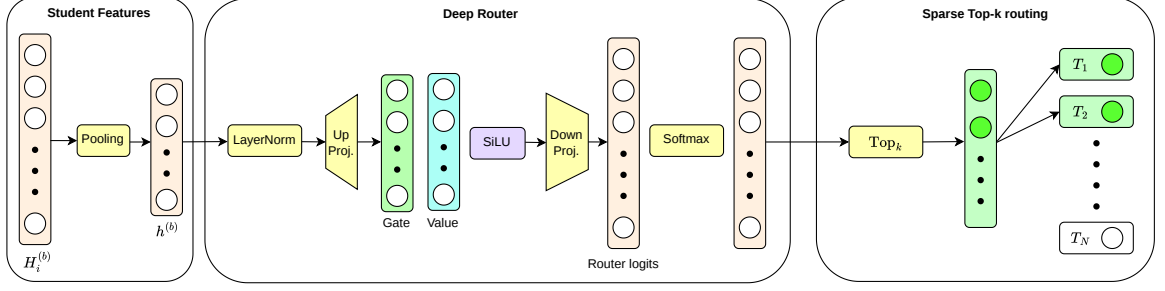


Figure 3.4: Teacher routing network.

The Byte-Trie Wasserstein loss performs exact transport under the trie metric. The complexity analysis below describes the cost of evaluating this sparse per-edge formula during training.

Complexity Analysis. The shared byte trie is built once with cost $O(V_s \bar{L}_s + V_t \bar{L}_t)$, where V_s, V_t are the student and teacher vocabulary sizes and $\bar{L}_s, \bar{L}_t$ are average trie-path lengths. During training, each aligned answer position keeps only the top- K_{TW} student and teacher tokens and accumulates probability mass along their active trie paths. The per-position cost is

$$O((V_s + V_t) \log K_{\text{TW}} + K_{\text{TW}}(\bar{L}_s + \bar{L}_t) + A),$$

where A is the number of active trie edges. Thus, sparse Byte-Trie Wasserstein matching avoids materializing the dense $|V_s| \times |V_t|$ optimal-transport matrix.

3.3 Teacher Routing Network

Uniform teacher averaging ignores how each teacher performs across different samples. GRACE therefore uses a student-conditioned router to propose a small teacher set for each sample before gradient-agreement refinement adjusts their weights. The full pipeline from pooled student features to sparse top- k teacher selection is shown in Figure 3.4 [25, 41, 42].

Let $\mathbf{H}_s^{(b)}$ be the student hidden states before the language-model head. GRACE pools the supervised answer positions to obtain the routing feature

$$\mathbf{h}^{(b)} = \text{Pool}(\mathbf{H}_s^{(b)}).$$

Answer-position pooling restricts the routing feature to the supervised span of the sequence, so the router sees a representation in which visual context, question text and answer states have already interacted. Teacher usefulness depends on the answer span being generated, not only on the input type.

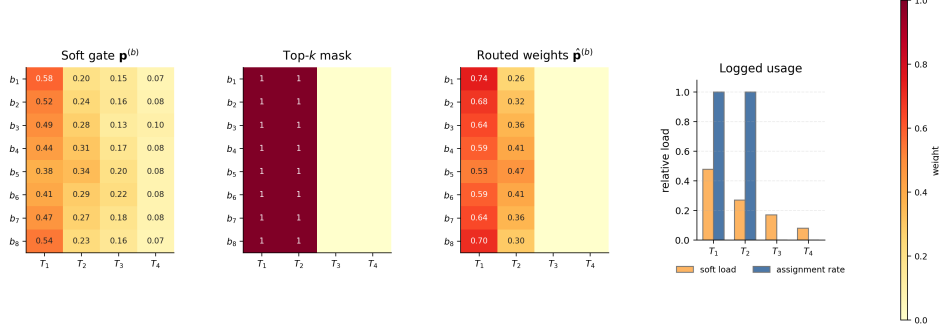


Figure 3.5: Top- k routing dynamics.

A learned MLP router $G_\phi(\cdot)$ maps this feature to teacher logits and soft weights:

$$\mathbf{r}^{(b)} = G_\phi(\mathbf{h}^{(b)}), \quad \mathbf{p}^{(b)} = \text{softmax}(\mathbf{r}^{(b)}).$$

The router then keeps the top- k_{route} teachers:

$$\mathcal{R}_b = \text{TopK}(\mathbf{r}^{(b)}, \min(k_{\text{route}}, N)).$$

Non-selected weights are zeroed and the selected weights are renormalized:

$$\hat{p}_i^{(b)} = \frac{\mathbf{1}[i \in \mathcal{R}_b] p_i^{(b)}}{\sum_{j=1}^N \mathbf{1}[j \in \mathcal{R}_b] p_j^{(b)}}. \quad (3.8)$$

The router determines which frozen teachers contribute to each student update without modifying the student architecture. Because $\mathbf{h}^{(b)}$ reflects the current student hidden states, the routed set $\mathcal{R}_b$ changes as the student evolves, so teacher selection adapts to both input content and the current student state.

Top- k routing has two roles. First, it makes the teacher mixture sparse: only k_{route} teacher losses can enter routed KD regardless of pool size. Second, it separates candidate selection from final weighting. The router proposes a candidate teacher set; Section 3.4 then describes how gradient agreement refines their weights. Router probability captures input-level relevance; gradient agreement checks whether the selected KD gradients support the supervised objective.

Figure 3.5 contrasts dense teacher probabilities with the sparse routed assignment.

Two auxiliary terms regularize teacher selection. The entropy term is a negative-entropy regularizer that discourages premature routing collapse [41, 42]:

$$\mathcal{L}_{\text{ent}} = -\frac{1}{B} \sum_{b=1}^B H(\mathbf{p}^{(b)}). \quad (3.9)$$

The z -loss penalizes large router logit magnitudes:

$$\mathcal{L}_z = \frac{1}{B} \sum_{b=1}^B \left(\log \sum_{i=1}^N e^{r_i^{(b)}} \right)^2. \quad (3.10)$$

The two terms serve different purposes. The entropy term discourages the router from assigning all samples to one teacher early in training, while the z -loss limits excessively sharp routing logits. These auxiliary losses stabilize teacher selection; CE and routed KD remain the main training signals.

3.4 Gradient-Agreement Refinement

Routing proposes teachers relevant to each sample, but a selected teacher may still produce a KD gradient that conflicts with the supervised CE objective. GRACE computes one agreement score per teacher per mini-batch and uses it to reweight each sample’s routed teachers.

3.4.1 Agreement Score Computation

Let $\mathbf{g}_{CE} = \nabla_{\Theta} \mathcal{L}_{CE}$ be the supervised gradient over trainable student parameters Θ . For teacher T_i , GRACE first averages its KD loss over the mini-batch and then computes

$$\mathbf{g}_i = \nabla_{\Theta} \left(\frac{1}{B} \sum_{b=1}^B \mathcal{L}_{KD}^{(b,i)} \right),$$

where $\mathcal{L}_{KD}^{(b,i)}$ is the Byte-Trie Wasserstein loss from Equation 3.7. The raw agreement score is the cosine similarity

$$s_i = \frac{\langle \mathbf{g}_{CE}, \mathbf{g}_i \rangle}{\max(\|\mathbf{g}_{CE}\|_2 \|\mathbf{g}_i\|_2, \epsilon_{\text{num}})}. \quad (3.11)$$

Cosine similarity is appropriate here because both the KD and CE losses optimize the same student parameters; it also normalizes for gradient magnitude, making the score independent of teacher-to-teacher scale differences. Consider one gradient step with routed KD:

$$\Delta \theta = -\eta \left(\mathbf{g}_{CE} + \alpha_{\text{KD}} \sum_i w_i \mathbf{g}_i \right).$$

The first-order change in the supervised CE loss is

$$\Delta \mathcal{L}_{CE} \approx \mathbf{g}_{CE}^{\top} \Delta \theta = -\eta \|\mathbf{g}_{CE}\|_2^2 - \eta \alpha_{\text{KD}} \sum_i w_i \langle \mathbf{g}_{CE}, \mathbf{g}_i \rangle.$$

When $\langle \mathbf{g}_{CE}, \mathbf{g}_i \rangle > 0$, teacher T_i ’s KD direction supports the supervised CE update. When the inner product is negative, the teacher pushes against the supervised CE objective. GRACE weights each routed teacher’s contribution by gradient agreement [47].

The score s_i is batch-level, computed once per teacher, but produces per-sample variation through its intersection with each sample’s routed set $\mathcal{R}_b$.

3.4.2 Warmup, EMA Smoothing and Active Set

GRACE converts raw agreement scores into final weights in four steps. During warmup (the first $\rho_g T_{\text{train}}$ steps), routed weights pass through unchanged: $w_i^{(b)} = \hat{p}_i^{(b)}$. Skipping agreement scoring early in training reflects a stability choice: before the router has seen enough data, agreement scores computed from an unstable student are too noisy to filter reliably. After warmup, the steps are as follows. First, the raw score is thresholded:

$$\tilde{a}_i = \mathbf{1}[s_i > \delta].$$

Second, it is smoothed with EMA:

$$\hat{s}_i = \beta_{\text{EMA}} \hat{s}_i^{\text{prev}} + (1 - \beta_{\text{EMA}}) s_i.$$

A noisy batch agreement score fluctuates across training steps; EMA dampens this before softmax weighting, as Figure 3.6 shows.

The raw score and smoothed score serve different purposes: s_i thresholds by current-batch agreement, while $\hat{s}_i$ weights teachers to smooth out per-batch noise.

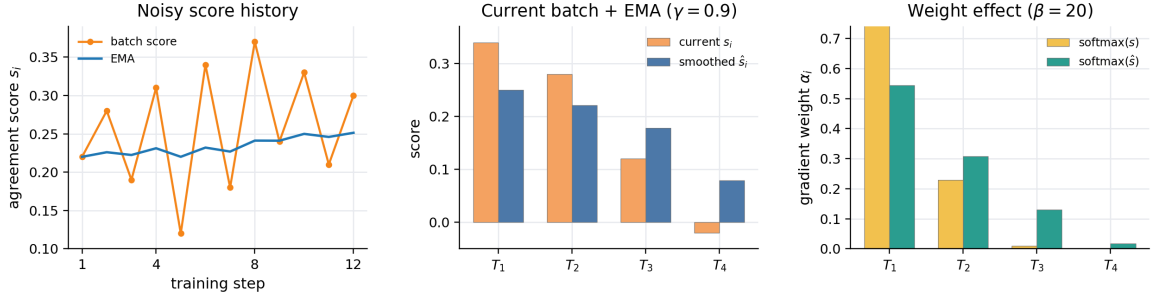


Figure 3.6: EMA smoothing.

Third, the active set for sample b is formed by intersecting the threshold flag $\tilde{a}_i$ with the routed set:

$$\mathcal{A}_b = \{i \in \mathcal{R}_b : \tilde{a}_i = 1\}. \quad (3.12)$$

Fourth, the smoothed scores are passed through a softmax over $\mathcal{A}_b$ to produce agreement weights ω :

$$\omega_i^{(b)} = \frac{\exp(\beta_{\text{score}} \hat{s}_i) \mathbf{1}[i \in \mathcal{A}_b]}{\sum_{j=1}^N \exp(\beta_{\text{score}} \hat{s}_j) \mathbf{1}[j \in \mathcal{A}_b]}, \quad (3.13)$$

3.4.3 Fallback, Uniform and Blended Weighting

The final rule, summarized in Figure 3.7, chooses among three cases. If $\mathcal{A}_b$ is empty, GRACE falls back to the routed weights $\hat{p}^{(b)}$. If the score spread inside $\mathcal{A}_b$ is below ϵ_{fb}

(scores nearly tied), GRACE uses uniform weights over $\mathcal{A}_b$. Otherwise, it geometrically blends the routed weights $\hat{p}$ and the agreement weights ω :

$$w_i^{(b)} = \frac{(\hat{p}_i^{(b)})^{\lambda_{\text{blend}}} (\omega_i^{(b)})^{1-\lambda_{\text{blend}}} \mathbf{1}[i \in \mathcal{A}_b]}{\sum_{j=1}^N (\hat{p}_j^{(b)})^{\lambda_{\text{blend}}} (\omega_j^{(b)})^{1-\lambda_{\text{blend}}} \mathbf{1}[j \in \mathcal{A}_b]}. \quad (3.14)$$

Here, $\lambda_{\text{blend}} \in [0, 1]$ interpolates between pure score weighting and pure router weighting.

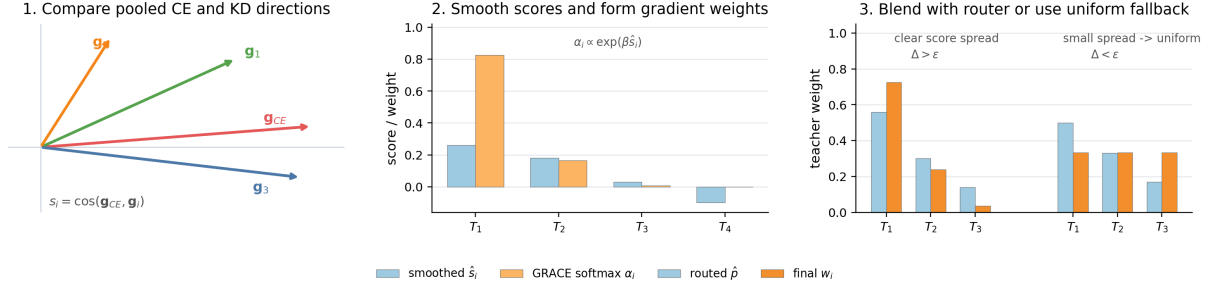


Figure 3.7: GRACE score refinement.

Figure 3.8 illustrates these three weighting outcomes after routing. When one active teacher is strongly preferred, the final weight shifts toward it. If the router and gradient agreement favor different teachers, the geometric blend preserves contributions from both. If active teacher scores are nearly equal, GRACE treats them as indistinguishable and uses uniform weights over the active set.

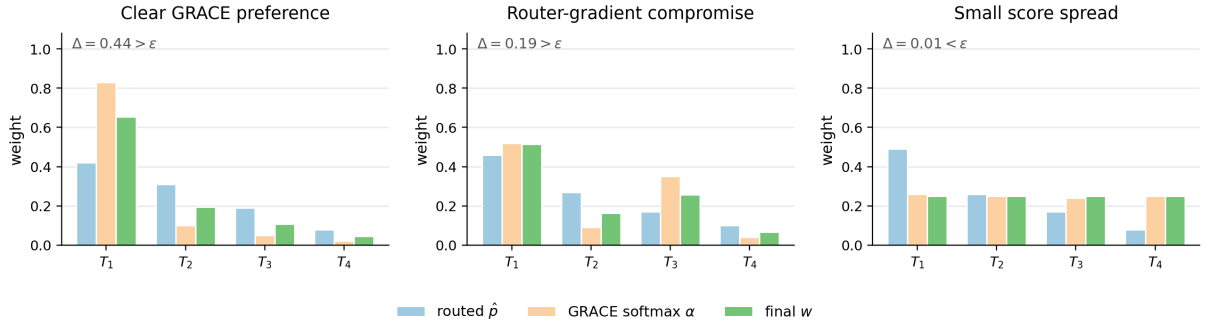


Figure 3.8: Representative GRACE weighting cases.

Figure 3.9 traces the per-sample weight path from dense router probabilities through top- k selection, agreement weights ω and final blending.

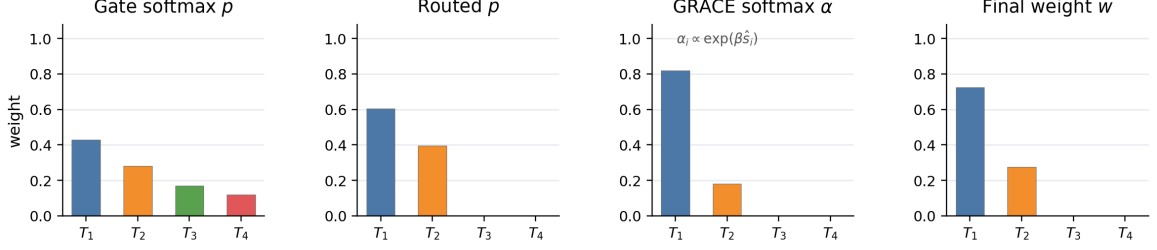


Figure 3.9: Sample-wise weight refinement.

Gradient-agreement refinement restricts the final mixture to the routed set $\mathcal{R}_b$, so the final routed KD term contains at most k_{route} nonzero teachers per sample. The per-teacher losses are computed before GRACE weights are applied, so the weighting step has access to all $\mathcal{L}_{KD}^{(b,i)}$ values.

3.5 Routed KD Aggregation and Final Objective

At the sample level, only teachers in the routed set $\mathcal{R}_b$ contribute to the KD loss:

$$\bar{\mathcal{L}}_{KD}^{(b)} = \sum_{i \in \mathcal{R}_b} w_i^{(b)} \mathcal{L}_{KD}^{(b,i)}, \quad (3.15)$$

where $w_i^{(b)}$ reflects whichever of the three weighting rules applies (fallback, uniform or blended). The mini-batch KD term averages these:

$$\mathcal{L}_{KD} = B^{-1} \sum_{b=1}^B \bar{\mathcal{L}}_{KD}^{(b)}.$$

Because $w_i^{(b)}$ is computed per sample, different examples in the same mini-batch can emphasize different teachers. Figure 3.10 illustrates this behavior for text-heavy, chart-heavy and ambiguous samples.

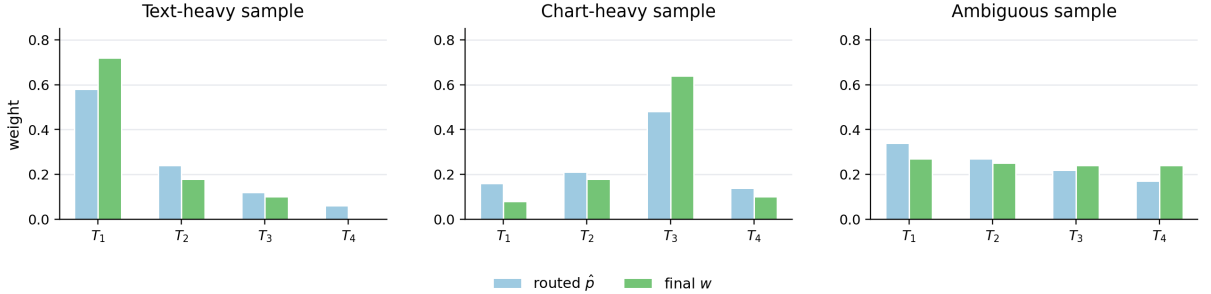


Figure 3.10: Sample-specific weighting.

Because each per-teacher loss uses Byte-Trie Wasserstein matching, this term stays valid across heterogeneous tokenizers without requiring aligned token indices.

The complete GRACE objective is

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha_{KD} \mathcal{L}_{KD} + \lambda_{ent} \mathcal{L}_{ent} + \lambda_z \mathcal{L}_z, \quad (3.16)$$

where $\mathcal{L}_{CE}$ anchors the student to supervised answer tokens, $\mathcal{L}_{KD}$ transfers routed teacher supervision and $\mathcal{L}_{ent}$ together with $\mathcal{L}_z$ regularize the router. Only θ and ϕ are updated.

Algorithm 1 One GRACE training step

- Require:** Mini-batch $\mathcal{B} = \{(x_b, y_b)\}_{b=1}^B$, frozen teachers $\{T_i\}_{i=1}^N$, student S_θ , router G_ϕ
- 1: Run the student on $\mathcal{B}$ to obtain logits $\mathbf{z}_s^{(b)}$, hidden states $\mathbf{H}_s^{(b)}$ and supervised loss $\mathcal{L}_{CE}$
 - 2: **for** each sample b **do**
 - 3: Pool hidden states at answer positions to form $\mathbf{h}^{(b)}$
 - 4: Compute router logits $\mathbf{r}^{(b)} = G_\phi(\mathbf{h}^{(b)})$ and probabilities $\mathbf{p}^{(b)} = \text{softmax}(\mathbf{r}^{(b)})$
 - 5: Select routed set $\mathcal{R}_b = \text{TopK}(\mathbf{r}^{(b)}, \min(k_{\text{route}}, N))$
 - 6: Renormalize routed weights to obtain $\hat{\mathbf{p}}^{(b)}$
 - 7: **end for**
 - 8: **for** each teacher T_i selected by at least one sample **do**
 - 9: Load cached teacher logits $\mathbf{z}_t^{(b,i)}$
 - 10: Compute $\mathcal{L}_{KD}^{(b,i)}$ with Byte-Trie Wasserstein matching on aligned answer positions
 - 11: Compute teacher-level gradient $\mathbf{g}_i$ and agreement score s_i
 - 12: **end for**
 - 13: Smooth agreement scores with EMA and construct active sets $\mathcal{A}_b$
 - 14: Compute final teacher weights $w_i^{(b)}$ using fallback, uniform or geometric blend rules
 - 15: Aggregate routed KD for each sample

$$\bar{\mathcal{L}}_{KD}^{(b)} = \sum_{i \in \mathcal{R}_b} w_i^{(b)} \mathcal{L}_{KD}^{(b,i)}$$

- 16: Optimize the final objective

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha_{KD} \frac{1}{B} \sum_{b=1}^B \bar{\mathcal{L}}_{KD}^{(b)} + \lambda_{ent} \mathcal{L}_{ent} + \lambda_z \mathcal{L}_z$$

Chapter 4

Evaluation Methodology

4.1 Experimental Setup

All experiments follow a shared evaluation protocol. The student models are **SmolVLM-500M** and **SmolVLM-256M** [20], assessed with **VLM-Eval-Kit** [50], which provides a shared generation-based inference, post-processing and metric-computation pipeline for all three benchmarks. For each benchmark sample, the toolkit loads the image-question pair, applies the dataset-specific prompt construction logic and calls the model’s generation interface to produce a free-form text answer.

Before training, all benchmark samples are converted into a unified **LLaVA-style multimodal instruction format** [3] as shown in listing 4.1, so TextVQA, DocVQA and ChartQA can share the same data loader, prompt template and answer supervision interface.

```
1 {  
2   "image": "sample_000123.jpg",  
3   "conversations": [  
4     {"from": "human", "value": "<image>\nWhat is the invoice number?"},  
5     {"from": "gpt", "value": "4572"}  
6   ]  
7 }
```

Listing 4.1: Representative LLaVA-style sample used after dataset conversion.

We run all training jobs on a single high-memory **NVIDIA H200** GPU or on comparable single-GPU **Vast.ai** instances, subject to availability. Distillation training uses **DeepSpeed ZeRO Stage 2**, **bf16** mixed precision, **TF32** matrix multiplication and gradient checkpointing to control memory use while keeping the student fully trainable and all teachers frozen [51, 52]. Full optimization and routing parameter values are reported in Appendix B, Table B.1. All reported scores reflect a single completed run per configuration; cost constraints precluded repeated runs with multiple random seeds. The implications of this design are discussed as a limitation in the Conclusion. Teacher logit storage and remote caching are documented in Appendix B.2 and Appendix B.3 for reproducibility purposes.



Figure 4.1: Representative dataset examples.¹

4.2 Datasets

We evaluate three text-centric VQA benchmarks: **TextVQA**, **DocVQA** and **ChartQA**. These benchmarks require reading scene text, parsing document layouts and reasoning over charts, covering complementary visual domains with OCR-heavy, layout-based and numerical demands [4–6].

TextVQA requires answering questions on natural images containing scene text, so the model must read and reason over text embedded in real-world scenes [4]. *Sample question*: “What word is written on the stop sign?”

DocVQA targets question answering on scanned document images, where layout, paragraphs, tables and forms are central to correct answers [5]. *Sample question*: “What is the invoice number shown in this document?”

ChartQA assesses question answering over charts, combining chart reading with numerical and logical reasoning [6]. *Sample question*: “Which year has the highest sales value in the chart?”

No single benchmark captures all three challenges; evaluating on all three prevents a method from appearing competitive through specialization on one task type alone.

Figure 4.1 shows representative samples from each dataset. The TextVQA panel shows a stop sign; the answer requires reading text embedded in the scene. The DocVQA panel shows a receipt with fields, item rows and totals arranged by document layout. The ChartQA panel shows a bar chart; answering requires comparing bar heights and reading values from the plot. Table 4.1 lists the split sizes.

¹Images adapted from Wikimedia Commons sources [53–55].

Table 4.1: Dataset split sizes.

Dataset	Train	Val	Test	Total
TextVQA	34,602	5,000	5,734	45,336
DocVQA	39,463	5,349	5,188	50,000
ChartQA-H (human)	7,398	960	1,250	9,608
ChartQA-M (machine)	20,901	960	1,250	23,111
ChartQA (H+M, combined)	28,299	1,920	2,500	32,719

The official metrics are dataset-specific: TextVQA uses VQA Accuracy, DocVQA uses ANLS and ChartQA uses Relaxed Accuracy. For TextVQA, the per-question score for a predicted answer $\hat{a}$ is $s_{\text{VQA}} = \min(\#\text{annotators that provided } \hat{a}/3, 1)$, and the final score is the mean over all questions [4]. For DocVQA, the metric is [5]:

$$\text{NLS}(a, \hat{a}) = 1 - \frac{d_{\text{Lev}}(a, \hat{a})}{\max(|a|, |\hat{a}|)}, \quad s_{\text{ANLS}} = \begin{cases} \text{NLS}(a, \hat{a}), & \text{NLS} \geq 0.5, \\ 0, & \text{otherwise.} \end{cases}$$

For ChartQA, we report Relaxed Accuracy [6]. Text predictions must match exactly, while numeric predictions are scored with

$$s_{\text{ChartQA}} = \begin{cases} \mathbf{1}[\hat{y} = y], & y = 0, \\ \mathbf{1}\left[\frac{|\hat{y}-y|}{|y|} \leq 0.05\right], & y \neq 0. \end{cases}$$

4.3 Baselines

Table 4.2 organizes the compared methods into four groups: non-distillation references, same-family KD, single-teacher transfer with mismatched tokenizers and heterogeneous teacher methods.

The model sources for the teacher setup column are SmolVLM2 [20, 56], Qwen2-VL [28, 57], Qwen2.5-VL [9, 58], InternVL3 [59] and Gemma 3 [60]. The non-distillation references consist of the original student and the full fine-tuning baseline. The same-family baselines use **SmolVLM2-2.2B-Instruct** as teacher for forward KL, reverse KL and Jensen-Shannon distillation, isolating standard teacher-student transfer when the tokenizer and vocabulary are aligned [16, 34, 61]. The different-family single-teacher setting uses ULD with Qwen2-VL-2B-Instruct outside the SmolVLM family, testing transfer across mismatched tokenizers with one teacher [21]. The heterogeneous teacher setting compares uniform averaging, SHM-CKA, REINFORCE and GRACE over a mixed teacher pool, separating static averaging, hidden-state matching and adaptive teacher assignment [22, 26, 27, 36].

Table 4.2: Baseline groups.

Group	Methods	Teacher setup	Purpose
Non-distillation references	Original, full fine-tuning	No teacher	Student-only reference points
Same-family baselines	Forward KL, Reverse KL, Jensen-Shannon	SmolVLM2-2.2B-Instruct	Standard single-teacher objectives within the SmolVLM family
Cross-family single-teacher	ULD	Qwen2-VL-2B-Instruct	Cross-family transfer with a single teacher
Heterogeneous teacher baselines	Uniform averaging, SHM-CKA, REINFORCE, GRACE (Ours)	Mixed teacher set from multiple VLM families	Static averaging vs. hidden-state matching vs. adaptive teacher assignment

All trainable competitor methods retain the supervised term $\mathcal{L}_{CE}$, so their differences come from the added transfer signal: the logits-only baselines change only the KD term, SHM-CKA adds hidden-state alignment and REINFORCE adds a policy-gradient term [24, 25, 62]. In the descriptions below, γ denotes a generic KD mixing coefficient, while α_{KD} is reserved for the GRACE objective in Equation 3.16.

Original. Evaluates the released student checkpoint, measuring the model’s score before fine-tuning or distillation.

Full fine-tuning. Uses supervised CE as the sole objective, which isolates task adaptation from any teacher-guided distillation signal.

Forward KL. Uses $\mathcal{L}_{KD} = \text{KL}(p_t||p_s)$ for same-family soft-target distillation when teacher and student vocabularies are aligned [16].

Reverse KL. Uses $\mathcal{L}_{KD} = \text{KL}(p_s||p_t)$ [61, 63] as a student-centered divergence baseline, which tests whether matching the teacher from the student’s current distribution is suitable for VLM distillation. Because the reverse-KL gradient is weighted by p_s , this objective tends to emphasize regions where the student already assigns probability.

Jensen-Shannon. Uses the symmetric divergence

$$\frac{1}{2}\text{KL}(p_s||m) + \frac{1}{2}\text{KL}(p_t||m), \quad m = \frac{1}{2}(p_s + p_t).$$

Jensen-Shannon tests whether a bounded symmetric objective is more stable than KL [34,

61].

ULD. Uses $\mathcal{L}_{KD} = \|\pi(p_s) - \pi(p_t)\|_1$ [21], where $\pi(\cdot)$ sorts the softened probability vector in descending order and pads unequal vocabulary sizes before comparison.

SHM-CKA. Extends ULD with a hidden-state CKA penalty [21, 24, 36]. For the heterogeneous setting, both the ULD and layer-matching terms are averaged uniformly across teachers before combining with $\mathcal{L}_{CE}$. Full implementation details are given in Appendix B.1.

Uniform multi-teacher. Assigns fixed equal weights to all teachers in the heterogeneous pool, $\mathcal{L}_{KD} = \frac{1}{M} \sum_{i=1}^M \mathcal{L}_{KD}^{(i)}$ [25, 26].

REINFORCE. Replaces deterministic routing with a stochastic policy that samples teachers and updates the selector with REINFORCE [25, 62], providing an adaptive routing baseline that replaces gradient-agreement refinement with policy-gradient selection. The selector is trained with the policy-gradient term

$$\mathcal{L}_{\text{pol}} = -(r - b) \log p(\mathbf{m} \mid \boldsymbol{\pi}), \quad (4.1)$$

where r is the reward, b is an EMA reward baseline and $\mathbf{m}$ is the sampled teacher mask. Its final training objective is

$$\mathcal{L}_{\text{RS}} = \mathcal{L}_{CE} + \gamma \mathcal{L}_{KD} + \lambda_{\pi} \mathcal{L}_{\text{pol}}. \quad (4.2)$$

Additional baseline implementation details are reported in Appendix B.1.

4.4 Ablation Settings

The ablation study in Section 5.2 isolates the main components of GRACE on DocVQA while keeping the teacher set and training protocol consistent with the full GRACE configuration. The variants change one mechanism at a time so the result table separates the contributions of teacher diversity, sparse routing, gradient-agreement weighting and EMA score smoothing.

Cross-family ULD. This variant uses one teacher outside the SmolVLM family and optimizes

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha_{\text{KD}} \mathcal{L}_{ULD}, \quad \mathcal{L}_{ULD} = \|\pi(p_s) - \pi(p_t)\|_1.$$

Uniform averaging. This variant adds the full heterogeneous teacher pool but assigns fixed equal weights:

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha_{\text{KD}} \frac{1}{M} \sum_{i=1}^M \mathcal{L}_{TW}^{(i)}.$$

This isolates teacher diversity from adaptive weighting.

Routing only. This variant keeps sparse teacher selection but replaces the router weights with uniform weights within the routed set:

$$w_i^{(b)} = u_i^{(b)} = \frac{\mathbf{1}[i \in \mathcal{R}_b]}{|\mathcal{R}_b|}.$$

This setting tests top- k routing before adding gradient-agreement refinement.

Full GRACE. This setting combines trie-Wasserstein KD, sparse routing and gradient-agreement weighting:

$$w_i^{(b)} \propto (\hat{p}_i^{(b)})^{\lambda_{\text{blend}}} (\omega_i^{(b)})^{1-\lambda_{\text{blend}}}.$$

Two additional ablation variants isolate the final weighting mechanism, reported in Table 5.5. The $\lambda_{\text{blend}} = 1.0$ variant keeps the threshold on the active set and fallback rules but removes the softmax over agreement scores from the high-confidence case:

$$w_i^{(b)} \propto \hat{p}_i^{(b)} \mathbf{1}[i \in \mathcal{A}_b].$$

Unlike Routing only, which weights all routed teachers, this variant restricts weights to the active set $\mathcal{A}_b$. The $\beta_{\text{EMA}} = 0$ variant removes temporal smoothing by setting $\hat{s}_i = s_i$ before softmax weighting over agreement scores.

Chapter 5

Experiment Results

Results address the three research questions from Chapter 1: whether GRACE outperforms same-family, cross-family and heterogeneous baselines; which components account for the gains; and what training overhead the method adds.

5.1 Main Comparison Results

We report the original performance of the two student models, **SmolVLM-500M** and **SmolVLM-256M** [20], using **VLM-Eval-Kit** [50] under the evaluation pipeline described in Chapter 4. Table 5.1 reports these student-only starting points together with the teacher reference range used in the teacher-gap comparisons.

Table 5.1: Student and teacher model results.

Model	TextVQA	DocVQA	ChartQA
<i>Student Models</i>			
SmolVLM-500M	56.75	59.52	62.80
SmolVLM-256M	47.17	48.64	45.04
<i>Teacher Models</i>			
Qwen2.5-VL-3B-Instruct	79.30	93.90	94.24
InternVL3-1B	74.10	81.90	75.30
Gemma-3-4B-it	57.80	75.80	63.60
Qwen2-VL-2B-Instruct	79.70	90.10	89.28

Model sources for Table 5.1 are SmolVLM [20], Qwen2.5-VL [9, 58], InternVL3 [59], Gemma 3 [60] and Qwen2-VL [28, 57].

Using the highest teacher score per benchmark as reference, the teacher-student gap reaches up to 34.38 points for SmolVLM-500M and 49.20 points for SmolVLM-256M, establishing the ceiling for these comparisons.

Table 5.2 collects the main results across four groups: (1) non-distillation references (original student and full fine-tuning), (2) same-family logit-based KD using **SmolVLM2-2.2B-**

Instruct [20, 56] as teacher (forward KL, reverse KL, JSD), (3) cross-family single-teacher ULD and (4) multi-teacher methods (Uniform, SHM-CKA, REINFORCE and GRACE), all using the same four-teacher set spanning the Qwen, InternVL and Gemma families [9, 28, 57–60]. Each cell reports the best completed run.

Table 5.2: Main comparison across TextVQA, DocVQA and ChartQA.

Category	Method	Teacher	Student	TextVQA	DocVQA	ChartQA
Non-distillation references						
Original	None	None	SmolVLM-500M	56.75	59.52	62.80
			SmolVLM-256M	47.17	48.64	45.04
Reference	Fine-tuning	None	SmolVLM-500M	56.96	61.96	63.00
			SmolVLM-256M	49.83	42.03	49.68
Same-family distillation baselines						
Same-family	Forward KL	SmolVLM2-2.2B	SmolVLM-500M	59.00	63.55	64.52
			SmolVLM-256M	50.88	49.46	56.80
	Reverse KL		SmolVLM-500M	44.73	29.28	30.16
			SmolVLM-256M	29.75	23.28	17.20
	Jensen-Shannon		SmolVLM-500M	42.33	36.43	43.75
			SmolVLM-256M	33.84	33.12	50.93
Different-family single-teacher baselines						
Cross-family	ULD	Qwen2-VL-2B-Instruct	SmolVLM-500M	61.49	65.28	66.92
			SmolVLM-256M	53.95	52.24	60.96
Multi-teacher baselines						
Multi-teacher	Uniform	†	SmolVLM-500M	63.88	69.59	75.60
			SmolVLM-256M	54.57	55.70	63.68
	SHM-CKA		SmolVLM-500M	60.47	63.32	76.88
			SmolVLM-256M	55.93	57.24	62.40
	REINFORCE		SmolVLM-500M	63.26	64.00	70.27
			SmolVLM-256M	56.31	58.53	68.74
	GRACE (Ours)		SmolVLM-500M	69.38	73.11	80.20
			SmolVLM-256M	60.73	60.48	72.32

† Teacher pool: Qwen2-VL-2B-Instruct, Qwen2.5-VL-3B-Instruct, InternVL3-1B, Gemma-3-4B-it; model sources cited above.

GRACE is the top method for both student sizes across all three benchmarks. Fine-tuning barely improves over the original student scores, and same-family logit-based KD depends strongly on the divergence: forward KL improves slightly, while reverse KL and Jensen-Shannon often reduce performance. Cross-family ULD outperforms same-family KD, and uniform multi-teacher averaging improves further, confirming that heterogeneous teachers add information beyond same-family transfer. Compared with the strongest non-GRACE baseline, GRACE improves by +5.50, +3.52 and +3.32 points for SmolVLM-

500M, and by +4.42, +1.95 and +3.58 points for SmolVLM-256M on TextVQA, DocVQA and ChartQA respectively. REINFORCE is the closest structural competitor, since it also routes adaptively, yet the 5.50-point gap on TextVQA for SmolVLM-500M indicates that gradient cosine similarity is a sharper teacher filter than a reward signal updated per evaluation checkpoint. Section 5.4 examines Reverse KL and SHM-CKA in more detail.

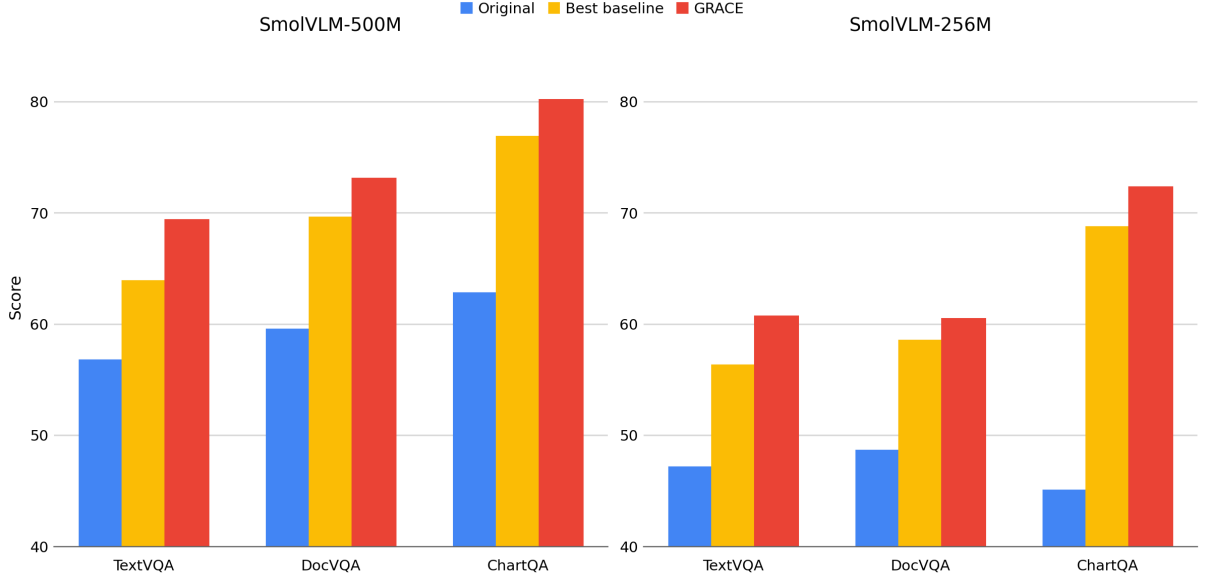


Figure 5.1: Grouped benchmark comparison.

Answer to RQ1

GRACE improves over direct fine-tuning, same-family logit-based KD, ULD and the compared heterogeneous multi-teacher baselines in Table 5.2. The score order is same-family logits < cross-family ULD < most multi-teacher settings < GRACE. Within the multi-teacher group, GRACE scores above Uniform, SHM-CKA and REINFORCE for both student sizes across the three benchmarks. Section 5.2 then examines which GRACE components account for these gains.

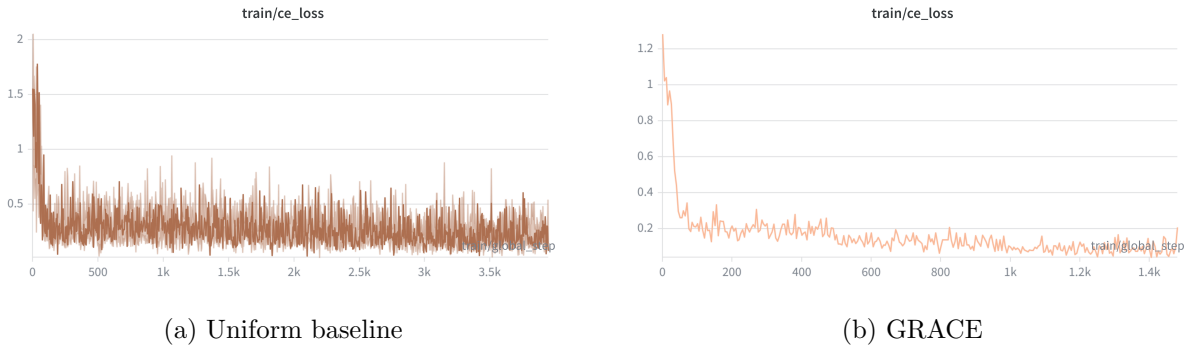
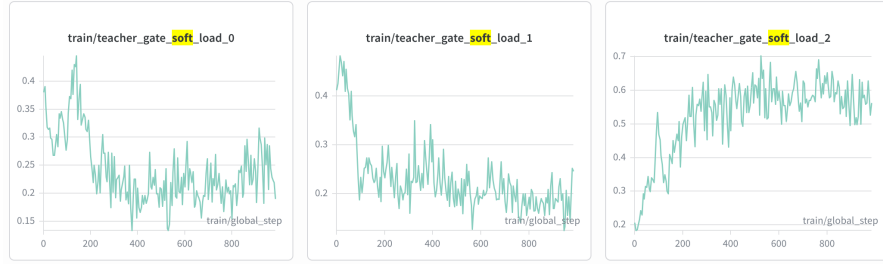


Figure 5.2: Uniform and GRACE training curves.

Figure 5.2 shows that the GRACE run has fewer sharp oscillations than Uniform on DocVQA.



(a) Without GRACE refinement



(b) With GRACE

Figure 5.3: Routed-teacher load examples.

Figure 5.3 contrasts router collapse with distributed teacher selection. Without the full GRACE pipeline, the routed load concentrates on one teacher. With GRACE, the load stays distributed, so the student receives heterogeneous supervision throughout. Table 5.3 quantifies these gains. Relative gain is computed as $(\text{GRACE} - \text{baseline})/\text{baseline}$.

Table 5.3: Relative gains of GRACE.

Benchmark	vs Original	vs Fine-tuning	vs Uniform	vs Highest non-GRACE
<i>SmolVLM-500M</i>				
TextVQA	+12.63 (+22.26%)	+12.42 (+21.80%)	+5.50 (+8.61%)	+5.38 (+8.41%)
DocVQA	+13.59 (+22.83%)	+11.15 (+18.00%)	+3.52 (+5.06%)	+3.52 (+5.06%)
ChartQA	+17.40 (+27.71%)	+17.20 (+27.30%)	+4.60 (+6.08%)	+3.32 (+4.32%)
<i>SmolVLM-256M</i>				
TextVQA	+13.56 (+28.75%)	+10.90 (+21.88%)	+6.16 (+11.29%)	+4.42 (+7.85%)
DocVQA	+11.84 (+24.34%)	+18.45 (+43.90%)	+4.78 (+8.58%)	+1.95 (+3.33%)
ChartQA	+27.28 (+60.57%)	+22.64 (+45.57%)	+8.64 (+13.57%)	+3.58 (+5.21%)

As Table 5.3 shows, gains over the original SmolVLM-500M student exceed 22% on TextVQA and DocVQA and 27% on ChartQA, with similar margins over direct fine-tuning. The gain over static uniform averaging is smaller but consistent, isolating the benefit of adaptive weighting over uniform teacher combination. The same trend holds for SmolVLM-256M, where gains reach +28.75%, +24.34% and +60.57%.

For the teacher-gap analysis, the reference point is the highest-scoring teacher in the GRACE comparison pool reported in Table 5.1:

$$g_{\text{orig}} = y_T - y_{\text{orig}}, \quad g_{\text{grace}} = y_T - y_{\text{grace}}.$$

The fraction of the gap closed is $1 - g_{\text{grace}}/g_{\text{orig}}$.

Table 5.4: Gap-to-teacher analysis for SmolVLM-500M.

Benchmark	Highest teacher	Original gap	GRACE gap	Gap closed
TextVQA	79.70	22.95	10.32	55.03%
DocVQA	93.90	34.38	20.79	39.53%
ChartQA	94.24	31.44	14.04	55.34%

GRACE reduces the gap to the strongest teacher. For SmolVLM-500M, GRACE closes 55.03%, 39.53% and 55.34% of the original teacher-student gap on TextVQA, DocVQA and ChartQA. Using the strongest available teacher as the ceiling makes this gap estimate conservative.

The distilled student can outperform individual teachers. On TextVQA, GRACE reaches 69.38 and exceeds Gemma-3-4B-it at 57.80 by +11.58 points (+20.03%). On ChartQA, it exceeds InternVL3-1B at 75.30 by +4.90 points (+6.51%) and Gemma-3-4B-it at 63.60 by +16.60 points (+26.10%). The routed mixture therefore draws on each teacher’s

strengths, occasionally surpassing individual teachers on specific benchmarks.

5.2 Ablation Study

This section answers RQ2 from Chapter 1 by isolating the main components of GRACE on DocVQA. Section 4.4 defines the ablation variants; results for both student models are in Table 5.5.

Table 5.5: DocVQA component ablation.

DocVQA ablation variant	SmolVLM	
	256M	500M
Cross-family ULD (single teacher)	52.24	65.28
Uniform averaging + trie-Wasserstein	55.70	69.59
Routing only + trie-Wasserstein	57.45	63.23
Full GRACE + trie-Wasserstein	60.48	73.11
w/o EMA ($\beta_{\text{EMA}} = 0$)	57.45	71.25
w/ fixed blend ($\lambda_{\text{blend}} = 1.0$)	57.97	70.45

The ablations compare single-teacher ULD, uniform multi-teacher KD, routing-only KD and full GRACE to separate teacher diversity, routing and gradient-agreement effects. Table 5.5 shows that full GRACE gives the highest DocVQA score for both SmolVLM-256M and SmolVLM-500M. Moving from single-teacher ULD to Uniform improves DocVQA by +3.46 and +4.31 points, confirming the value of heterogeneous teacher diversity. Routing only behaves inconsistently across student sizes: it improves SmolVLM-256M by +1.75 points over Uniform but reduces SmolVLM-500M by -6.36 points. The drop for SmolVLM-500M is, in our view, the most telling result of the ablation: a router that selects teachers without checking gradient alignment can be worse than no routing at all. Gradient-agreement refinement addresses this: full GRACE improves over Routing only by +3.03 points for SmolVLM-256M and +9.88 points for SmolVLM-500M. Removing EMA lowers DocVQA from 60.48 to 57.45 for SmolVLM-256M and from 73.11 to 71.25 for SmolVLM-500M. Setting $\lambda_{\text{blend}} = 1.0$ also decreases both scores, so both EMA smoothing and score-based blending contribute to peak performance.

Answer to RQ2

The ablation shows that heterogeneous teacher diversity improves over single-teacher ULD, while routing behaves inconsistently before gradient-agreement refinement is added. The largest contribution comes from gradient-agreement refinement, which accounts for the margins just reported. EMA smoothing and blending by agreement scores further improve final performance.

5.3 Efficiency Analysis

This section answers RQ3 by summarizing the training overhead introduced by GRACE. The main online costs come from three components: trie-Wasserstein support construction, teacher routing and gradient-agreement scoring. Trie-Wasserstein matching builds sparse token supports and accumulates trie-edge masses at answer positions. The router is an MLP over pooled student states; its cost scales with the hidden size and number of teachers. Gradient-agreement scoring introduces overhead from comparing teacher-specific KD gradients, bounded by top- k routing.

Teacher inference is offline: cached logits remove the cost of running frozen teachers during training, shifting that computation to a one-time caching step. The main scaling factors are the number of teachers, the retained trie support size K_{TW} and the number of routed teachers k_{route} . An average training run on the reported configurations costs approximately **14.41 USD**, excluding one-time offline teacher logit caching and remote storage. This figure covers the student forward and backward passes, sparse routing computation and gradient-agreement scoring for the full training run.

Answer to RQ3

GRACE adds overhead from trie-Wasserstein matching, routing and gradient-agreement scoring, while offline caching replaces repeated teacher forward passes with logit loading from cache. Per answer position, trie-Wasserstein matching scales with K_{TW} , routing scales with the number of teachers N , and top- k routing keeps the active teacher count bounded by k_{route} . Thus, the method trades per-step matching and scoring cost for distillation with offline pre-cached teacher logits and sparse teacher selection at each step.

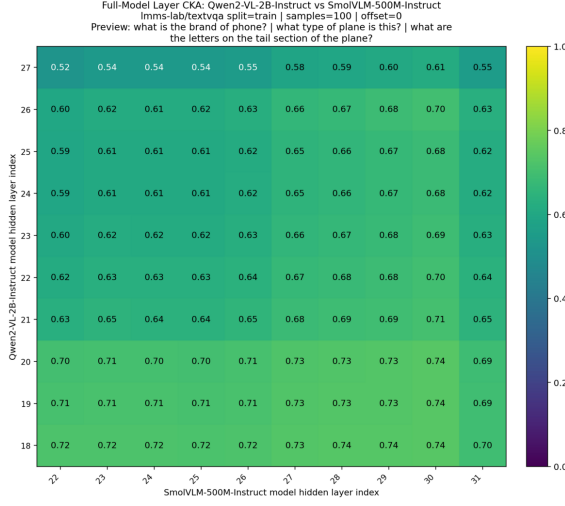
5.4 Discussion

The results show that heterogeneous distillation requires more than adding teachers to the loss: the gains come from combining cross-tokenizer output matching, sample-dependent routing and gradient reweighting after routing [16, 21, 38, 42]. Figure 5.2 shows fewer oscillations for GRACE than Uniform on DocVQA, consistent with restricting reweighting to the routed teacher subset.

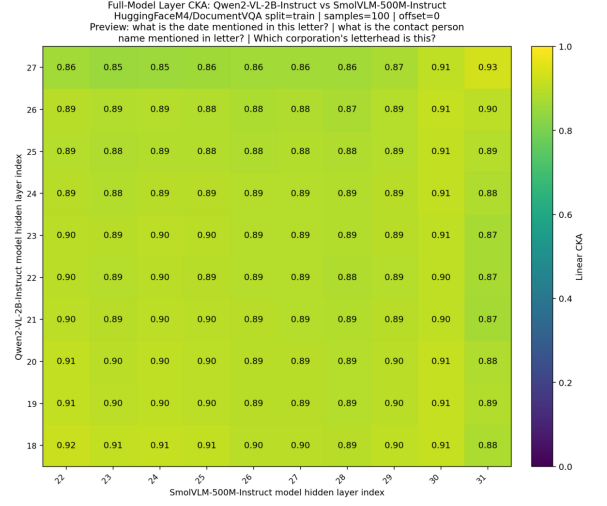
Behavior of same-family logit-based KD. The same-family logit-based baselines show different behavior across distributional losses for distillation on answer tokens. Forward KL gives small but consistent gains, while Reverse KL performs substantially worse in this setting. A likely reason is that Reverse KL is weighted by the current student distribution, so tokens the teacher assigns high probability to but that the student currently assigns low probability receive weaker correction. This is undesirable for OCR-heavy and numerically sensitive VQA tasks, where teacher probability mass may appear on low-probability answer variants. Reverse KL is, in short, the wrong objective when the student must recover teacher probability mass on tokens it currently neglects. This explanation is limited to the reported KD experiments on answer tokens and should not be read as a general claim about Reverse KL in all distillation settings [64].

Cross-family ULD. ULD consistently outperforms all same-family KD variants despite using only one teacher. The likely reason is that sorted-rank matching avoids the shared-vocabulary assumption entirely: comparing sorted probability vectors penalizes distributional shape differences without requiring that corresponding indices refer to the same token. ULD therefore sets the cross-family baseline that Uniform and GRACE must exceed. However, rank-based comparison discards the semantic relationship between tokens: two tokens with adjacent ranks but unrelated decoded strings are treated as equally close. Byte-Trie Wasserstein replaces rank proximity with byte-path proximity, giving lower transport cost to tokens with similar decoded content. This is why GRACE improves over ULD even with the same Qwen2-VL teacher in the pool.

Representation matching and SHM-CKA. SHM-CKA tests whether hidden-state alignment alone transfers knowledge across heterogeneous teachers. It is competitive on ChartQA for SmolVLM-500M but trails GRACE overall [24, 37].

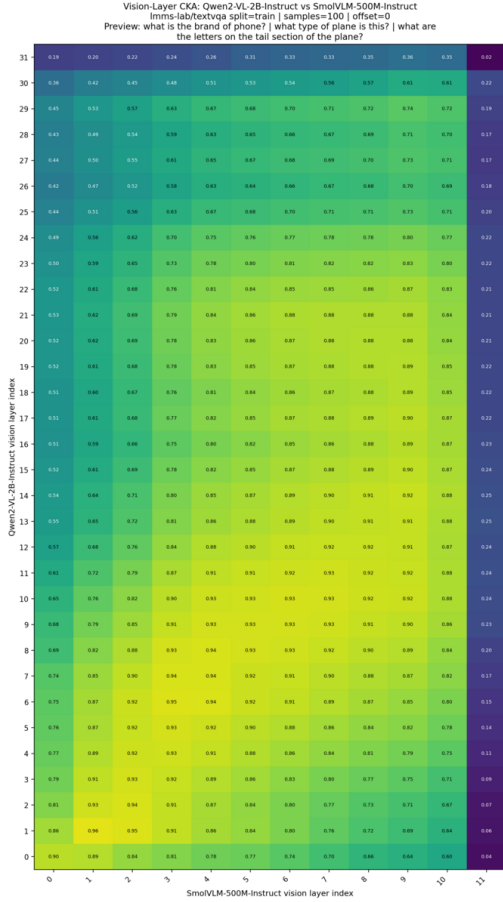


(a) TextVQA

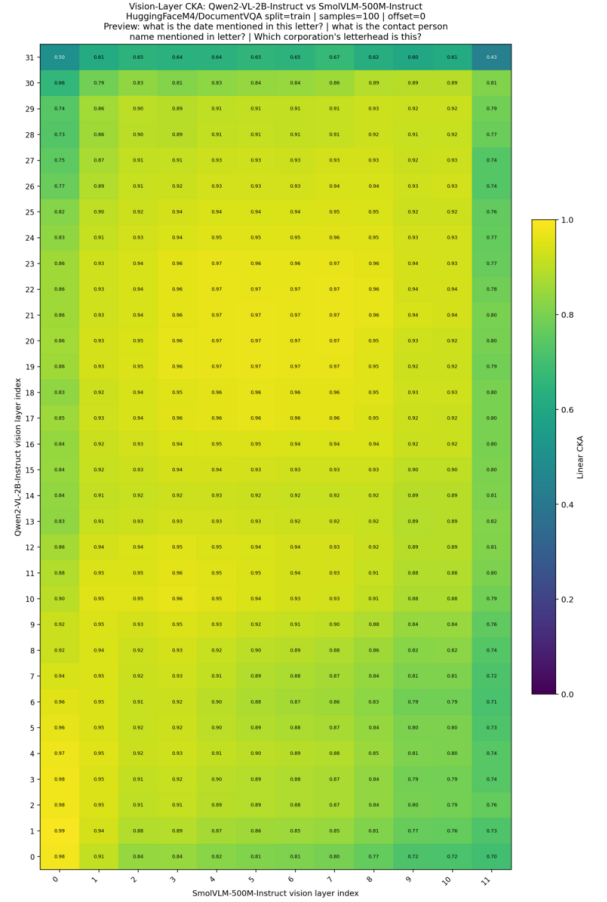


(b) DocVQA

Figure 5.4: Final-layer CKA heatmaps.



(a) TextVQA



(b) DocVQA

Figure 5.5: Vision-layer CKA heatmaps.

Figures 5.4 and 5.5 show higher CKA values on DocVQA than TextVQA, especially across late language layers and middle vision layers. Nevertheless, Table 5.2 shows that SHM-CKA

trails output-level distillation in several settings, confirming that high CKA does not guarantee strong answer prediction. We read this as evidence that representation alignment and output calibration are largely orthogonal objectives in heterogeneous VLM distillation.

First-order interpretation of gradient-agreement refinement. The first-order analysis in Section 3.4 shows that a teacher whose KD gradient conflicts with the CE gradient partially offsets the supervised update. Suppressing such teachers explains the ablation finding that gradient-agreement refinement outperforms routing alone, consistent with treating routing as candidate selection and gradient agreement as corrective reweighting [47].

Visual grounding. To inspect whether distillation changes which image regions the student attends to, we use DEX-AR attribution maps [65], which aggregate token-level visual attributions into a sequence-level heatmap for autoregressive VLMs. Figure 5.6 shows a representative TextVQA example [4] where the question asks “*who is this copyrighted by?*” and the relevant evidence is the copyright text on the lower-left border. The original SmolVLM-500M concentrates on the *Virgin* logo; GRACE redirects attention to the copyright text, correctly locating the region needed to answer the question. The shift follows from the output-distribution supervision: pushing the student toward teacher probability mass on the correct answer tokens causes the model to attend to the image regions that support those tokens.

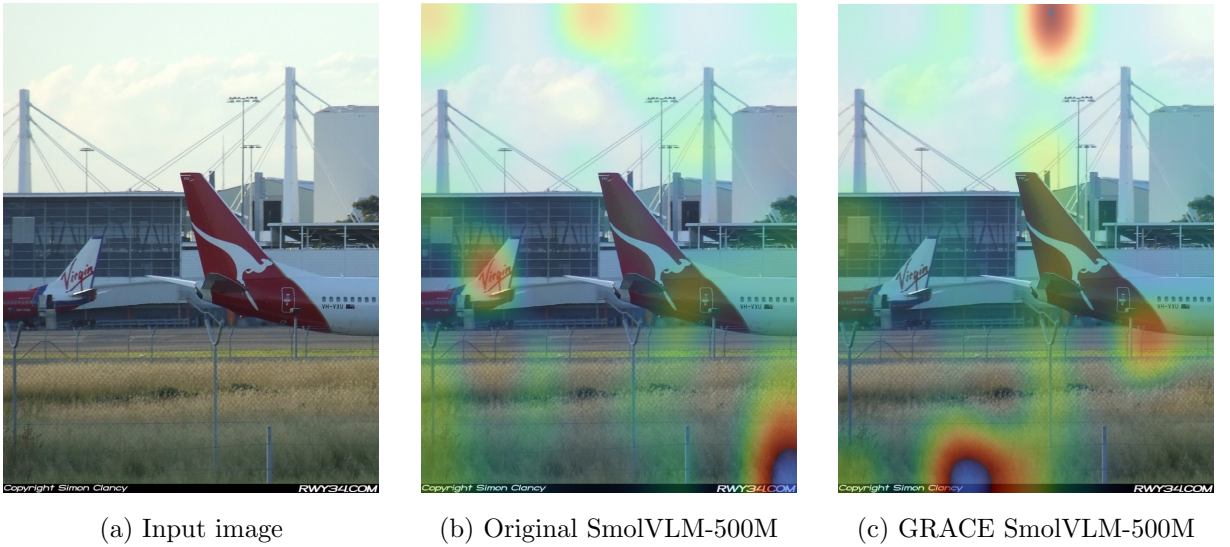


Figure 5.6: Representative TextVQA attention example.

Conclusion and Future Work

Summary

This thesis studied knowledge distillation for heterogeneous vision-language models where teachers vary in usefulness across samples and differ from the student in tokenizer. Building on cross-tokenizer distillation, optimal transport and sparse routing [21, 38, 42], the work introduced Gradient-Routed Alignment with Capacity-aware Experts (GRACE), a routed multi-teacher framework that combines three main contributions: Byte-Trie Wasserstein tokenizer-agnostic distillation, sparse top- k teacher routing and gradient-agreement refinement.

Byte-Trie Wasserstein distillation maps decoded teacher and student tokens into a shared byte-trie structure and compares subtree probability masses across tokenizer vocabularies. The objective-aware refinement combines sparse routing, which selects teachers relevant to each sample, with gradient-agreement reweighting by the cosine alignment between each teacher’s KD gradient and the supervised answer gradient. The empirical evaluation on TextVQA, DocVQA and ChartQA shows that this combination outperforms same-family, single-teacher and heterogeneous multi-teacher baselines.

Across TextVQA, DocVQA and ChartQA, GRACE scores highest among the compared methods in Chapter 5. For SmolVLM-500M, it reaches 69.38 on TextVQA, 73.11 on DocVQA and 80.20 on ChartQA. For SmolVLM-256M, it reaches 60.73, 60.48 and 72.32 on the same three benchmarks. On SmolVLM-500M, GRACE closes about 55% of the original teacher-student gap on TextVQA and ChartQA and about 40% on DocVQA. Closing more than half the gap to teachers six times larger suggests that the supervision design, rather than model capacity, is the primary bottleneck for compact VLMs on these benchmarks.

Limitations

Three limitations qualify the current results. **Cost-limited evaluation.** The reported single-run design limits uncertainty analysis for small gaps between high-scoring baselines because confidence intervals and estimates from repeated runs are unavailable. As reported in Section 5.3, an average run costs approximately 14.41 USD before offline cache generation and storage cost. Although the main tables use a shared four-teacher pool and consistent hyperparameters, no uncertainty estimates are available for closely matched methods. Small score differences between high-scoring baselines should be treated with caution

until repeated-run estimates are available.

Scope of validation. The empirical study focuses on VQA benchmarks and SmolVLM-family students. The results support VQA distillation with compact projector-based students, while broader multimodal task families, other student architectures and larger teacher pools require further validation. Tasks with long-form answers, multi-image inputs, video reasoning or open-ended dialogue require different forms of teacher selection and answer supervision.

Reasoning-oriented model support remains outside the current scope. GRACE currently distills teacher supervision through output distributions at answer tokens and gradient agreement with the supervised answer loss. This design fits the evaluated VQA setting, where the target is a short final answer. It focuses on supervision of final answers and leaves intermediate deliberation traces, hidden reasoning tokens and extended test-time reasoning for future work. If a teacher produces only a final answer distribution, GRACE can still use that distribution as a teacher signal. Future extensions should separate tokens in the reasoning phase from those in the answer phase, route teachers by the quality of intermediate reasoning, and define distillation objectives for multi-step reasoning traces.

Future Work

Four directions remain. The most immediate gap is statistical: repeated runs with multiple random seeds would clarify which score differences between close multi-teacher baselines are reliable. On the systems side, more compact logit storage, adaptive cache loading and lower-cost agreement scoring would reduce the cost of scaling the method to larger datasets and teacher pools. Beyond output-level supervision, visual token alignment, cross-attention transfer and region-level matching could complement the current KD signal for tasks where the student must localize evidence before answering. Reasoning-oriented models raise a harder problem: distinguishing reasoning-phase tokens from final-answer tokens and specifying how teachers are selected when intermediate traces are available. Router input features, principled conflict resolution and teacher pools that combine specialized document, chart, general VLM and reasoning-oriented models are additional open questions [25–27].

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Appendix A

Proof of the Closed-Form Trie-Wasserstein

This appendix derives the closed-form expression used by the KD loss in Chapter 3. The result is stated for a general rooted tree [22, 38].

The proof uses the following notation. The tree $T = (V, E, r)$ is a rooted tree with root r and each edge $e \in E$ has weight $w(e) > 0$. For an edge $e = (p(e), c(e))$, define the subtree below e as

$$\Gamma(e) = \{v \in V \mid e \text{ lies on the path from } r \text{ to } v\}.$$

For vertices $u, v \in V$, the tree metric is

$$d_T(u, v) = \sum_{e \in \text{path}(u, v)} w(e).$$

μ and ν are probability measures on V . In the trie-Wasserstein setting, these are the student and teacher token distributions on the leaves of the shared byte trie, with zero mass assigned to non-leaf vertices. The Kantorovich form of the W_1 distance is

$$W_{1,T}(\mu, \nu) = \min_{\pi \in \Pi(\mu, \nu)} \sum_{u, v \in V} \pi(u, v) d_T(u, v),$$

where $\Pi(\mu, \nu)$ is the set of couplings satisfying

$$\sum_{v \in V} \pi(u, v) = \mu(u), \quad \sum_{u \in V} \pi(u, v) = \nu(v), \quad \pi(u, v) \geq 0.$$

For each edge e , the subtree imbalance is

$$\Delta(e) = \mu(\Gamma(e)) - \nu(\Gamma(e)).$$

We prove that

$$W_{1,T}(\mu, \nu) = \sum_{e \in E} w(e) |\Delta(e)|.$$

We fix any feasible coupling $\pi \in \Pi(\mu, \nu)$ and any edge $e \in E$. Removing e separates the tree into two components: $\Gamma(e)$ and $V \setminus \Gamma(e)$. Any mass transported from one side of this cut to the other must cross e .

The outward and inward crossing masses are

$$q_\pi^{\text{out}}(e) = \sum_{u \in \Gamma(e)} \sum_{v \notin \Gamma(e)} \pi(u, v), \quad q_\pi^{\text{in}}(e) = \sum_{u \notin \Gamma(e)} \sum_{v \in \Gamma(e)} \pi(u, v).$$

Thus the total mass crossing e is $q_\pi(e) = q_\pi^{\text{out}}(e) + q_\pi^{\text{in}}(e)$.

The marginal constraints imply that the source mass in $\Gamma(e)$ is

$$\mu(\Gamma(e)) = \sum_{u \in \Gamma(e)} \sum_{v \in \Gamma(e)} \pi(u, v) + q_\pi^{\text{out}}(e),$$

The target mass in $\Gamma(e)$ has the analogous decomposition

$$\nu(\Gamma(e)) = \sum_{u \in \Gamma(e)} \sum_{v \in \Gamma(e)} \pi(u, v) + q_\pi^{\text{in}}(e).$$

Subtracting gives the required cut imbalance:

$$\Delta(e) = q_\pi^{\text{out}}(e) - q_\pi^{\text{in}}(e), \quad |\Delta(e)| \leq q_\pi^{\text{out}}(e) + q_\pi^{\text{in}}(e) = q_\pi(e).$$

Thus every feasible transport plan must move at least $|\Delta(e)|$ mass across edge e .

Since the path between any two vertices in a tree is unique,

$$d_T(u, v) = \sum_{e \in E} w(e) \mathbf{1}\{e \in \text{path}(u, v)\}.$$

The cost of π therefore decomposes as

$$\sum_{u, v \in V} \pi(u, v) d_T(u, v) = \sum_{e \in E} w(e) q_\pi(e).$$

The inequality $q_\pi(e) \geq |\Delta(e)|$ gives

$$\sum_{u, v \in V} \pi(u, v) d_T(u, v) \geq \sum_{e \in E} w(e) |\Delta(e)|.$$

Minimizing over all feasible π gives

$$W_{1,T}(\mu, \nu) \geq \sum_{e \in E} w(e) |\Delta(e)|.$$

It remains to construct a feasible plan attaining this lower bound. Define the signed measure by

$$\sigma(v) = \mu(v) - \nu(v).$$

Since μ and ν are probability measures,

$$\sum_{v \in V} \sigma(v) = 0.$$

For each edge e , the imbalance can be written as

$$\Delta(e) = \sum_{v \in \Gamma(e)} \sigma(v).$$

Thus $\Delta(e)$ is the net surplus mass in the subtree below e .

The quantity $\Delta(e)$ can be interpreted as a signed flow across edge e . If $\Delta(e) > 0$, then $\Delta(e)$ units of surplus mass move upward from $\Gamma(e)$ toward the root. If $\Delta(e) < 0$, then $|\Delta(e)|$ units of mass move downward into $\Gamma(e)$. Because the total signed mass is zero, the total surplus equals the total deficit, so this flow can satisfy all demands. More explicitly, for each non-root vertex x , the edge to its parent is $e_x = (p(x), x)$, and the flow is

$$F(e_x) = \Delta(e_x).$$

The subtree below e_x consists of x and the subtrees below the child edges of x . Therefore,

$$F(e_x) = \sigma(x) + \sum_{y \in \text{Ch}(x)} F(e_y).$$

This identity expresses flow conservation: the net mass sent through the parent edge of x equals the local surplus at x plus the surplus received from its children. At the root, the same conservation condition follows from $\sum_{v \in V} \sigma(v) = 0$.

The signed edge flow F transports the positive part of σ to the negative part of σ . Since T is a tree, this edge flow can be decomposed into transport paths from surplus vertices to deficit vertices. π^* denotes the corresponding coupling. This coupling leaves common mass fixed at each vertex and the fixed mass has zero cost. At vertex v , this mass equals $\min\{\mu(v), \nu(v)\}$, so $\pi^* \in \Pi(\mu, \nu)$. By construction, the amount of mass transported across each edge e under π^* is exactly $|\Delta(e)|$. Therefore,

$$\sum_{u, v \in V} \pi^*(u, v) d_T(u, v) = \sum_{e \in E} w(e) |\Delta(e)|.$$

This construction gives the upper bound

$$W_{1,T}(\mu, \nu) \leq \sum_{e \in E} w(e) |\Delta(e)|.$$

Combining the lower and upper bounds yields

$$W_{1,T}(\mu, \nu) = \sum_{e \in E} w(e) |\mu(\Gamma(e)) - \nu(\Gamma(e))|.$$

Appendix B

Implementation Details

This appendix lists the optimization and routing hyperparameters used across all distillation experiments and records the engineering details needed to reproduce the training infrastructure. The hyperparameters correspond to the evaluation protocol described in Chapter 4.

Table B.1: Experiment parameters.

Setting	Value
<i>Model setup</i>	
Student model	HuggingFaceTB/SmolVLM-500M-Instruct [20]
Teacher models	Qwen/Qwen2.5-VL-3B-Instruct [9, 58], Qwen/Qwen2-VL-2B-Instruct [28, 57], OpenGVLab/InternVL3-1B [59], google/gemma-3-4b-it [60]
Teacher weighting strategy	routing (GRACE path)
Distillation loss	Trie-Wasserstein
Trie-Wasserstein rho	0.7
Trie-Wasserstein top- k	64
Validation split	validation
<i>Hardware and infrastructure</i>	
Hardware and runtime	Single-GPU H200 or Vast.ai runs
Distributed training	DeepSpeed ZeRO-2
Numerical precision	bf16 enabled, half precision (fp16) disabled, TF32 enabled
FlashAttention-2	Enabled
Training data format	LLaVA-style multimodal instruction JSON
<i>Training</i>	
Max sequence length	16384
Training epochs	2
Per-device batch size	80
Gradient accumulation steps	1
Effective batch size	80
Gradient checkpointing	Enabled
Frozen modules	None
Dataloader workers	4

Continued on the next page.

Table B.1 continued from the previous page.

Setting	Value
Logging and checkpointing	Log every 5 steps and save every 150 steps
Seed	42
<i>Optimizer</i>	
Optimizer	adamw_torch
Adam betas	(0.9, 0.98)
Adam epsilon	1×10^{-7}
Weight decay	0.0
Base learning rate	1×10^{-5}
Vision learning rate	2×10^{-6}
Connector learning rate	1×10^{-5}
Learning-rate schedule	Cosine decay with warmup ratio 0.03
<i>GRACE hyperparameters</i>	
Student/teacher temperature	$\tau_s = \tau_t = 1.0$
KD weight	$\alpha_{\text{KD}} = 0.5$
Top- k routing	$k_{\text{route}} = 2$, with top-scoring teacher masking and selected softmax-weight renormalization
Gate entropy weight	$\lambda_{\text{ent}} = 10^{-3}$
Router z -loss weight	$\lambda_z = 10^{-3}$
GRACE warmup ratio	$\rho_g = 0.05$
GRACE softmax beta	$\beta_{\text{score}} = 4.0$
GRACE blend exponent	$\lambda_{\text{blend}} = 0.9$
GRACE EMA decay	$\beta_{\text{EMA}} = 0.8$
Score-spread fallback threshold	$\epsilon_{\text{fb}} = 0.01$
Alignment threshold	$\delta = 0.15$

B.1 Baseline Implementation

This section records the additional baseline definitions used for the comparison methods in Chapter 4, including how their teacher selection, weighting and auxiliary loss terms are implemented for fair comparison.

SHM-CKA. The baseline first loads an offline CKA matrix and, for each student layer s , selects the top- k teacher layers $\mathcal{T}_s$. Their CKA scores are normalized into soft

weights $\alpha_{s,t}$. The per-teacher layer loss is

$$\mathcal{L}_{\text{layer}}^{(i)} = \frac{1}{|S|} \sum_{s \in S} \sum_{t \in \mathcal{T}_s} \alpha_{s,t} (1 - \sqrt{\text{CKA}(\mathbf{H}_s, \mathbf{H}_t)}). \quad (\text{B.1})$$

SHM-CKA also assigns each teacher a ULD logits loss. The averaged terms are

$$\bar{\mathcal{L}}_{\text{ULD}} = \frac{1}{M} \sum_{i=1}^M \mathcal{L}_{\text{ULD}}^{(i)}, \quad \bar{\mathcal{L}}_{\text{layer}} = \frac{1}{M} \sum_{i=1}^M \mathcal{L}_{\text{layer}}^{(i)}. \quad (\text{B.2})$$

The final SHM-CKA objective is

$$\mathcal{L}_{\text{SHM-CKA}} = \frac{1}{3} (\mathcal{L}_{CE} + \bar{\mathcal{L}}_{\text{ULD}} + \bar{\mathcal{L}}_{\text{layer}}). \quad (\text{B.3})$$

REINFORCE. The policy uses the pooled student hidden state together with per-teacher summary features. A linear policy predicts independent Bernoulli probabilities

$$\boldsymbol{\pi} = \sigma(W \text{LN}(\mathbf{z}) + b), \quad (\text{B.4})$$

then samples a teacher mask $m_i \sim \text{Bernoulli}(\pi_i)$. If no teacher is selected, the highest-probability teacher is used as fallback. The selected mask is normalized into teacher weights

$$w_i = \frac{m_i}{\sum_{j=1}^M m_j}, \quad \mathcal{L}_{KD} = \sum_{i=1}^M w_i \mathcal{L}_{KD}^{(i)}. \quad (\text{B.5})$$

The selector is trained with the policy-gradient term

$$\mathcal{L}_{\text{pol}} = -(r - b) \log p(\mathbf{m} \mid \boldsymbol{\pi}), \quad (\text{B.6})$$

where r is the reward and b is an EMA reward baseline. The total reinforced-selection objective is

$$\mathcal{L}_{\text{RS}} = \mathcal{L}_{CE} + \gamma \mathcal{L}_{KD} + \lambda_{\pi} \mathcal{L}_{\text{pol}}. \quad (\text{B.7})$$

B.2 Offline Teacher-Logit Preparation

This section records the offline teacher-logit generation step used before distillation training. An average full H200 student-training run takes about 7 hours and costs about 14.41 USD at 2.059 USD/hour on Vast.ai, excluding the separate cost of offline teacher-logit generation and remote storage.

Teacher logits are generated *offline* before distillation. An offline caching script runs each frozen teacher on every training example, keeps only the supervised answer-token positions

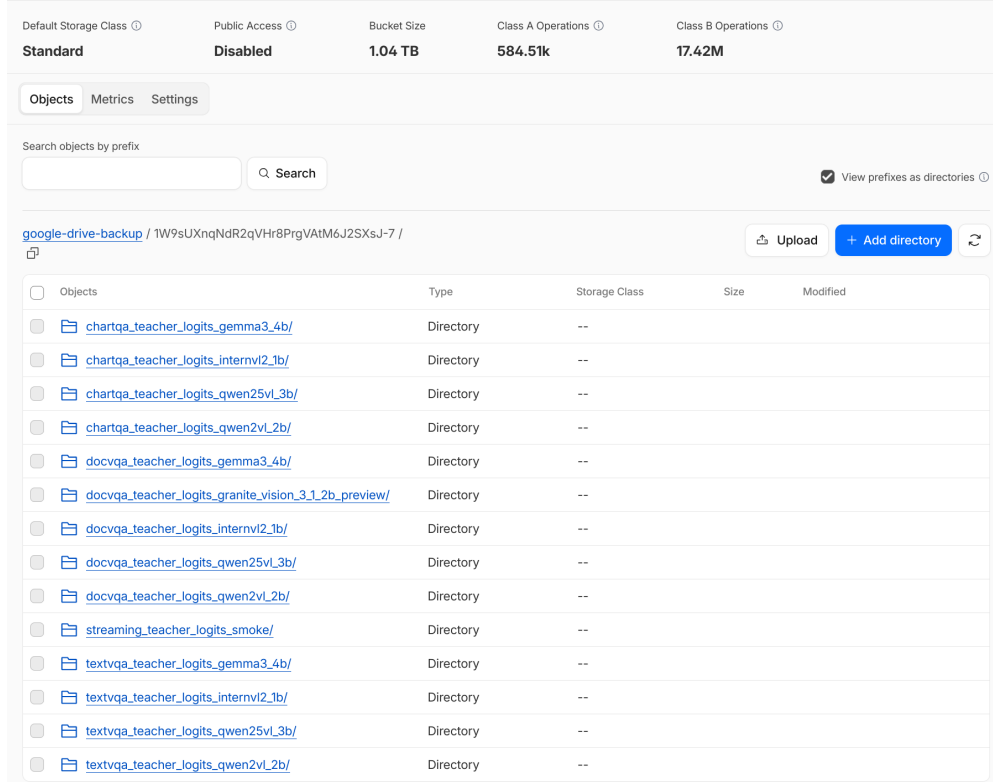


Figure B.1: R2 teacher-logit storage.

and writes one `.pt` file per sample under a teacher-specific directory. Each record stores the sample index, the selected teacher logits and the corresponding supervised labels. This keeps the online training loop simple: the student only reads precomputed teacher targets, so the teachers do not run again. After generation, the teacher-logit directories are uploaded to a **Cloudflare R2** bucket using its S3-compatible object-storage interface [66, 67]. In the active setup, the bucket already exceeds **1 TB**, so copying the full cache to an external GPU instance is not feasible. This is especially relevant on services such as Vast.ai, where local NVMe space is usually much smaller than the full teacher logit store. Figure B.1 shows this offline storage layout and highlights why the training path must stream teacher logits without staging the full cache locally.

B.3 Remote Logit Streaming During Training

The main training path does *not* download the full cache before training. Figure B.2 shows the full remote-cache path: teacher logits are generated offline and uploaded once, each training step requests a sample, the local cache serves hits directly and cache misses trigger an R2 download followed by least-recently-used eviction when the 100 GB limit is exceeded. When the launcher receives an R2 root in `distill_multi_teachers.sh`, the trainer switches to remote lazy loading. During training, `StreamingTeacherLogitsCache`

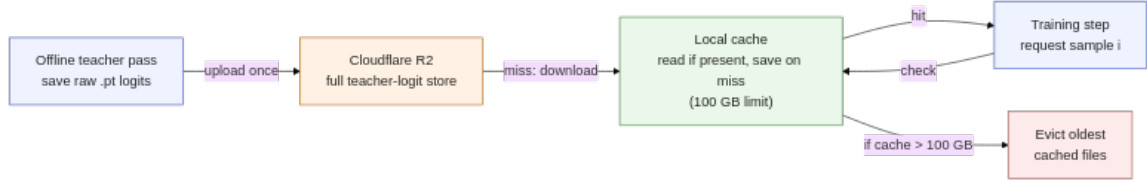


Figure B.2: Training with remote teacher logits.

resolves the teacher-specific remote object path for each requested sample, then fetches files lazily from R2 through `s3fs` and `WholeFileCacheFileSystem` [68, 69]. To keep disk usage bounded, the loader maintains a **100 GB** local cache. Each file is cached on first access, its access time is updated on each subsequent open and the oldest files are evicted once the cache exceeds the limit. The local cache therefore behaves as a small least-recently-used buffer over the much larger R2 store. A per-teacher file lock guards cache metadata updates so that concurrent dataloader workers do not corrupt the cache state.

Table B.2: Logit-streaming settings.

Setting	Value
Teacher-logit storage	Cloudflare R2 object storage
Teacher-logit format	Raw per-sample .pt files
Remote path template	<code>remote_root/</code> <code><dataset>_teacher_logits_</code> <code><alias>/<teacher-slug>/</code> <code><index>.pt</code> <code>/tmp/teacher-logits-</code> <code>streaming</code> for R2 reads
Local cache directory	<code>/workspace/cache</code> for local-cache reads
Local cache limit	100 GB